

APPH E3300 Applied Electromagnetism

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Abstract

Vector Analysis (Chapter 1) II. Electrostatics (Chapter 2) III. Solution Techniques for Electrostatic Problems Laplace's Equation (3.1) Method of Images (3.2) Multipole Expansions (3.4) IV. Electric Fields in Matter (Chapter 4) Polarization (4.1-4.3) Dielectric Materials (4.4) V. Application Examples of Electrostatics VI. Magnetostatics (Chapter 5) VII. Magnetic Fields in Matter (Chapter 6) Magnetization (6.1-6.4.1) Ferromagnetism (6.4.2) VIII. Application Examples of Magnetostatics

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Chapter 1

Overview of EM

Lecture 1

Review of Class Syllabus

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Chapter 2

Vector Calculus and Gauss's Law

Lecture 2

Remark. Differential Calculus: Gradient, Divergence

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Definition 2.0.1. Del operator

$$\nabla = i_x \frac{\partial}{\partial x} + i_y \frac{\partial}{\partial y} + i_z \frac{\partial}{\partial z}$$

The del operator alone is meaningless, however when used with scalar or vector fields, it gives rise to interesting concepts

Definition 2.0.2. We define Divergence as

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \sum_i^n \frac{\partial f(\vec{A})}{\partial A_i} \vec{e}_i$$

For a given divergence, we define the flux of a vector field. We define a surface with area dx, dy with normal vector $\hat{n}$. The flux differential is defined to be

$$d\phi = \hat{n} \cdot \vec{A} dx dy$$

For a large surface, the flux is defined to be

$$\phi = \int d\phi = \int \vec{A} \cdot \hat{n} ds$$

From here we get Gauss's divergence theorem (Green's Theorem)

$$\oint \vec{A} \cdot \hat{n} ds = \int_V (\nabla \cdot \vec{A}) dV$$

Given a uniform vector field, the flux is 0. Next we define the curl of the vector with is

$$\nabla \times \vec{A}$$

which defined how much the vector "swirls" around points. The circulation of a field is denoted as

$$C = \oint \vec{A} \cdot d\vec{l}$$

We will get the between the curl and the circulation to be Stoke's Theorem

$$\oint \vec{A} \cdot d\vec{l} = \int_s (\nabla \times \vec{A}) \cdot \vec{n} dS$$

Remark. Second Order Derivatives: scalar fields and vector fields

Gradient $\vec{\nabla} f$ which is vector. We can apply the del operator again with the dot product or the cross product which are two ways. For the divergence $\vec{\nabla} \cdot \vec{A}$ is applying the del operator again $\vec{\nabla}(\nabla A)$. For the curl, we have two combinations that are the same as the gradient. Thus, there are 5 second order derivatives...

Definition 2.0.3.

$$\nabla \cdot (\nabla f) = \sum_i^n \frac{\partial^2}{\partial x_i^2} f = \nabla^2 f$$

which is also defined as the laplacian

Definition 2.0.4.

$$\nabla \times (\nabla f) = \left(\frac{\partial^2}{\partial yz} - \frac{\partial^2}{\partial z,y}\right)\hat{x} + \left(\frac{\partial^2}{\partial z,x} - \frac{\partial^2}{\partial x,z}\right)\hat{y} + \left(\frac{\partial^2}{\partial x,y} - \frac{\partial^2}{\partial y,x}\right)\hat{z} = 0$$

Definition 2.0.5.

$$\nabla(\nabla \cdot A)$$

Definition 2.0.6.

$$\nabla \cdot (\nabla \times A) = 0$$

Definition 2.0.7.

$$\nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A$$

Remark. Curvilinear Coordinates:

There exists curvilinear coordinates which are useful in projection in orthogonal directions in 3D.

Example. The use of spherical and cylindrical coordinates are important for point charges, conducting wires, calculating the flux for certain special objects and planes, and other important objects. Spherical coordinates are useful in certain descriptions such as that of a point charge.

Definition 2.0.8. Spherical Coordinates:

Suppose we had a point p in spherical coordinates with coordinates (r, ϕ, θ) where the ranges are going to be

$$0 \leq r \leq \infty$$

$$0 \leq \phi \leq 2\pi$$

$$0 \leq \theta \leq \pi$$

where ϕ represents the movement in the x-y plane and θ represents rotation on the z-y plane. The unit vectors are going to follow in the same planes. The conversion is as follows for cartesian to spherical

$$x = r \cos \phi \sin \theta, y = r \sin \phi \sin \theta, z = r \cos \theta$$

The infinitesimal displacement will be

$$d\hat{l} = dr\hat{r} + r d\theta\hat{\theta} + r \sin \phi d\phi\hat{\phi}$$

For a small volume, we have

$$dV = r^2 \sin \theta dr d\phi d\theta, (dx dy dz)$$

Definition 2.0.9. Cylindrical Coordinates (r, ϕ, z)

$$0 \leq r \leq \infty, 0 \leq \phi \leq 2\pi, -\infty \leq z \leq \infty$$

infinitesimal displacement:

$$d\hat{l} = dr\hat{r} + dz\hat{z} + r d\phi\hat{\phi}$$

Volume:

$$dV = r dr d\phi dz$$

Lecture 3

Remark. Dirac delta function:

Consider a vector field $\vec{V} = \frac{1}{r^2}\hat{r}$. The divergence of the field will result to 0 which is surprising

$$\vec{\nabla} \cdot \vec{v} = \vec{\nabla} \cdot \left(\frac{1}{r^2} \hat{r} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = 0$$

For Gauss's Theorem, we have that

$$\int \vec{\nabla} \cdot \vec{v} = \oint \vec{v} \cdot \hat{n} dS = 4\pi$$

This 4π value holds for any radius, thus there exists a singularity which we can use to define the dirac delta function.

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Definition 2.0.10. Properties:

$$\begin{aligned}\delta(x) &= 0, x \neq 0 \\ \int_{-\infty}^{\infty} \delta(x) dx &= 1 \\ \int_{-\infty}^{\infty} \delta(x)f(x) dx &= f(0) \quad \int_{-\infty}^{\infty} \delta(x) dx = f(0) \\ \delta(x-a) &= 0, x \neq a \\ \int_{-\infty}^{\infty} f(x)\delta(x-a) dx &= f(a)\end{aligned}$$

There are ways to approximate $\delta(x)$ by having a rectangular series ranging from $\pm \frac{1}{2n}$

$$\delta_n(x), n \hookrightarrow \infty$$

Another example is the function

$$\delta_n(x) = \frac{n}{\sqrt{\pi}} e^{-n^2 x^2}$$

where we must have

$$\lim_{n \rightarrow \infty} \delta_n(x)f(x)dx = f(0)$$

and the normalization will still have that

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

Definition 2.0.11. 3D Dirac Delta function

$$\begin{aligned}\delta^3 &= 0, \vec{x} \neq 0 \\ \int_{-\infty}^{\infty} \delta(\vec{r} - \vec{r}_0) dV &= 0 \\ \int_{-\infty}^{\infty} \delta^3(\vec{r} - \vec{r}_0)f(\vec{r}) dV &= f(\vec{r}_0)\end{aligned}$$

The physical case of a delta function is going to have the form

$$\begin{aligned}\vec{E} &= \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2} \\ \vec{\nabla} \cdot \vec{E} &= \frac{q}{4\pi\epsilon_0} \left(\vec{\nabla} \cdot \frac{\hat{r}}{r^2} \right) = \frac{q}{\epsilon_0} \delta^3(\vec{r})\end{aligned}$$

where we later find that $q\delta^3(r)$ to be the charge density. Many times the derivative of the delta function is not well defined. However, when we place it in a integral it is intuitive...

$$\int_{-\infty}^{\infty} dx f(x) \frac{d\delta(x)}{dx} = -\frac{df(x)}{dx} \Big|_{x=0}$$

Theorem 2.0.1. Helmholtz's Theorem

Given $\vec{\nabla} \cdot \vec{F} = D$ and $\vec{\nabla} \times \vec{F} = \vec{c}$ where the divergence of the curl is 0. The theorem states that if $D, \vec{c} \hookrightarrow 0, \vec{r} \hookrightarrow \infty$ and we have the condition $\vec{F} \hookrightarrow 0, \vec{r} \hookrightarrow \infty$. We can write $\vec{F}$ as two parts.

$$\vec{F} \equiv -\vec{\nabla}\phi + \vec{\nabla} \times \vec{A}$$

where

$$\vec{\nabla} \cdot \vec{F} = -\nabla^2 \phi = D$$

$$\vec{\nabla} \times \vec{F} = \nabla \times (\vec{\nabla} \vec{A}) = \vec{c}$$

If $\vec{c} = 0$, $\vec{F}$ is irrotational or curless and is written as $\vec{F} = -\vec{\nabla} \phi$ where $\oint \vec{F} \cdot d\vec{l} = 0$ or also independent from the path. Another special case is when $D = 0$, we call the field $\vec{F}$ to be divergenceless or solenoidal where we write $\vec{F} = \vec{\nabla} \times \vec{A} \Rightarrow \oint \vec{F} \cdot d\vec{S} = 0$ which means that it is independent of the surface for any given boundary.

Remark. Electrostatics:

Suppose we had a point charge q_0, q_1 at positions $\vec{r}_0, \vec{r}_1$ where we can find that

$$\vec{F}_{01} = \frac{q_0 q_1}{4\pi\epsilon_0} \frac{\hat{r}_{01}}{|\vec{r}_{01}|^2}, \vec{r}_{01} = \vec{r}_0 - \vec{r}_1$$

Suppose we have q_0 fixed and move q_1 in multiple places, we will see that the force

$$\vec{F}_{01}(\vec{r}_0) = q_0 \frac{q_1}{4\pi\epsilon_0} \frac{\hat{r}_0}{|\vec{r}|^2}$$

to be the field generated by charge q_1 . Thus generally we have that

$$\vec{F} = q\vec{E}$$

For a general set of particles we have

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_j \frac{q_j(\vec{r} - \vec{r}_j)}{|\vec{r} - \vec{r}_j|^3}$$

We can then generalize for a continuous charge distribution where we look at an infinitesimal distribution of $dq = \rho dV$ to be

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}_1)(\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3} dV$$

where the charge density ρ is in units of $\frac{C}{m^3}$. We will later see for surface charges such as a charge sheet, the charge is distributed on the sheet with density σ , the infinitesimal charge will be $dq = \sigma dA$. For line charges, we can write for a small segment to be $dq = \lambda dl, \lambda = \frac{C}{m}$.

Example. Consider an $\vec{E}$ field of a circular loop of charge along the z -axis. We calculate that

$$dE_z = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dl}{R^2 + z^2} \cdot \frac{z}{\sqrt{R^2 + z^2}}$$

$$E_z = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{R^2 + z^2} \cdot \frac{z}{\sqrt{R^2 + z^2}} \int_0^{2\pi} dl$$

$$= \frac{R}{2\epsilon_0} \frac{\lambda}{R^2 + z^2} \cdot \frac{z}{\sqrt{R^2 + z^2}}$$

Example. Another example is calculating the electric field of an infinitely large charge sheet at a point z above the sheet. We can partition the sheet into a circle and see the symmetry that exists for rings. For a single ring we know the result. We can then integrate to get that

$$E_z = \frac{z\sigma}{2\epsilon_0} \int_0^\infty \frac{R}{(R^2 + z^2)^{\frac{3}{2}}} dR$$

$$E_z = \frac{z\sigma}{2\epsilon_0} \frac{1}{\sqrt{R^2 + z^2}} \Big|_0^\infty$$

$$E_z = \frac{\sigma}{2\epsilon_0}$$

Remark. Gauss's Law

$$\vec{E}(\vec{r}) = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2}$$

$$\oint_S \vec{E}(\vec{r}) \cdot \hat{n} dS = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{R^2} 4\pi R^2 = \frac{q}{\epsilon_0}$$

We get there that the total flux is going to be

$$Flux = \oint_S \vec{E} \cdot \hat{n} dS = \sum_j q_j / \epsilon_0$$

Lecture 4

2.1 Gauss's Law

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For an electric field of a point charge in space is given as

$$\vec{E}(\vec{r}) = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2}$$

Choosing a surface to be the sphere of radius r

$$\oint_s \vec{E}(\vec{r}) \cdot \hat{n} dS = \frac{q}{4\pi\epsilon_0} \frac{1}{R^2} 4\pi R^2 = \frac{q}{\epsilon_0}$$

We can generalize this to multiple point charges in space where we can find a closed surface around it. Thus we have that the total flux to be

$$Flux = \oint_S \vec{E} \cdot \hat{n} dS = \frac{\sum_i q_i}{\epsilon_0} = \frac{Q_{tot}}{\epsilon_0}$$

From Gauss law we have that

$$\oint_S \vec{E} \cdot \hat{n} dS = \int_V (\nabla \cdot \vec{E}) dV = \int_V \frac{\rho}{\epsilon_0} dV$$

this holds for any shape or surface. Thus, this tells us that

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

This is the differential shape of Gauss's law and the other is the integral form of Gauss's law...

Example. Suppose we had a sphere with uniform volume of charge ρ and we want to find the E field inside and outside the sphere. Note we have spherical symmetry with spherical coordinates. We can choose our Gauss's surface to be a sphere as well, choosing radius a we have two cases: For $r < a$

$$\oint_S |\vec{E}(\vec{r})| dS = |\vec{E}(\vec{r})| 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$= |E(\vec{r})| = \frac{r\rho}{3\epsilon_0}$$

$r \geq a$

$$\Rightarrow |E(\vec{r})| = \frac{\rho a^3}{3\epsilon_0 r^2}$$

Plotting the electric field as a function of r we have that we get a linear increase and we have a $\frac{1}{r^2}$ decay

Example. Suppose we have a charged sheet with surface charge σ . Now suppose we know Gauss's law then we can form an infinitesimal box with surface area dS , using Gauss's law we have that

$$\oint \vec{E} \cdot n dS = 2|\vec{E}| = \frac{\sigma A}{\epsilon_0}$$

note that we have two surfaces to consider since our surface goes through the whole sheet.

$$|\vec{E}| = \frac{\sigma}{2\epsilon_0}$$

Thus, this is a simple derivation of the E field.

Example. Consider the volume charge of a cylinder ρ with uniform charge density. The radius is given as r , then calculating we have

$$\vec{E} \parallel \hat{r}$$

then for $r < R$, we can create a smaller cylinder

$$E \cdot 2\pi r L = \frac{\rho}{\epsilon_0} \pi r^2 L$$

$$E = \frac{\rho}{2\epsilon_0} r$$

For $r > R$, we have

$$E 2\pi r L = \frac{\rho}{\epsilon_0} \pi R^2 L$$

$$\Rightarrow E = \frac{\rho R^2}{2\epsilon_0 r}$$

Remark. It is important to have symmetry to construct Gauss's law in order to determine the electric field easily.

2.2 Curl of E

Suppose we had a charge at the origin of the coordinate where we have that

$$\vec{E} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r}$$

Suppose we wanted to calculate the line integral from point a to point b along some path I . In spherical coordinates we have that

$$d\vec{l} = dr\hat{r} + r d\theta\hat{\theta} + r \sin\theta d\phi\hat{\phi}$$

Thus the line integral must be

$$\vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$$

$$\int_a^b \vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \int_a^b \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right)$$

No matter what path we choose, the path integral only depends on the end points, this implies that the field must be conservative. If we perform a integral around we have that

$$\oint_c \vec{E} \cdot d\vec{l} = 0 \Rightarrow \nabla \times \vec{E} = 0$$

Note we have discovered Maxwell's equations for the electrostatics case. Note that for the time dependent case where we write

$$\oint_c \vec{E} \cdot d\vec{l} = - \int_s \frac{\partial B}{\partial t} \cdot \hat{n} dS$$

The potential here will be ill-defined and the E and B field are a function of time.

2.3 Electric Potential

Recall Helmholtz theorem that we can redefine the electric field as

$$\vec{E} = -\nabla\Phi$$

Note that the minus sign is by convention. Observe Gauss's law

$$\nabla \cdot \vec{E} = \nabla \cdot (-\nabla\Phi) = -\nabla^2\Phi = \frac{\rho}{\epsilon_0}$$

$$\nabla^2\Phi = \frac{-\rho}{\epsilon_0}$$

which is known as Poisson's Equation. When we have that $\rho = 0$, this gives us Laplace's equation

$$\nabla^2\Phi = 0$$

Note the extra constraint we had where $\nabla \times \vec{E} = 0$ from Helmholtz Theorem where

$$\vec{F} = -\nabla\Phi + \nabla \times \vec{A} \Rightarrow \nabla \cdot \vec{F} = -\nabla^2\Phi, \nabla \times \vec{A} = \nabla \times \vec{F}$$

Observe the units where

$$|E| : \frac{N}{C} = \text{V/m}$$

$$\Phi : \frac{N \cdot m}{C} = \text{volt}$$

We can convert to integral form where

$$\int_a^b \vec{E} \cdot d\vec{l} = \int_a^b (-\nabla\Phi) \cdot d\vec{l} = -(\Phi(b) - \Phi(a))$$

Remark. We see the gradient theorem for the integral of the electric field which is also path independent. Note that we can add a scalar field s.t. $\Phi' = \Phi + C$ which would not affect the integral of the E field. Thus, we can choose a reference point where we can choose $r = \infty$ as our reference point and define the potential $\Phi(\infty) = 0$. Thus, calculating the potential at

$$\Phi(P) = - \int_{\infty}^P \vec{E} \cdot d\vec{l}$$

Example. Consider the example of a point particle at distance r away from the point charge, we have

$$\begin{aligned} \vec{E} &= \frac{q}{4\pi\epsilon_0 r^2} \hat{r} \\ \phi(P = R) &= - \int_{\infty}^P \vec{E} \cdot d\vec{l} \\ &= - \int_{\infty}^P \frac{q}{4\pi\epsilon_0 r^2} dr \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{R} \end{aligned}$$

Remark. We can generalize to a collection of charges for this idea of a potential.

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_j \frac{q_j}{|\vec{r} - \vec{r}_j|}$$

For continuous charge distributions, we denote

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV'$$

where all that changes is the symbol σ, ρ, λ , for surface, volume, and line charges. Thus, we see it is easier to work with potentials than electric fields due to the $\frac{1}{r}$ dependence rather than the $\frac{1}{r^2}$ dependence. Note that it is important to be careful on how you choose your points for your potential. For example, a charged sheet should be chosen s.t. one point is infinity away along the z-axis and not along the xy plane.

Example. Suppose we had a charged circular sheet with surface charge density σ , what is the potential along the z-axis. We can construct a thin ring. Therefore we have that

$$\begin{aligned} \Phi &= \frac{1}{4\pi\epsilon_0} \int_0^R \frac{2\pi a da \cdot \sigma}{\sqrt{z^2 + a^2}} \\ &= \frac{2\pi\sigma}{4\pi\epsilon_0} \left. \sqrt{z^2 + a^2} \right|_0^R \\ &= \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - |z|) \end{aligned}$$

We can find the electric field easily since

$$\vec{E} = -\nabla\Phi$$

In cartesian coordinates we have

$$\frac{\sigma}{2\epsilon_0} \left[1 \mp \frac{z}{\sqrt{z^2 + R^2}} \right] \hat{z}, \quad - \text{ for } z > 0$$

Thus, we see a discontinuity in the charged sheet in the electric field.

Lecture 5

Remark. Recall the integration technique for a point z away from a ring of charge q .

Feb 5th 10:10

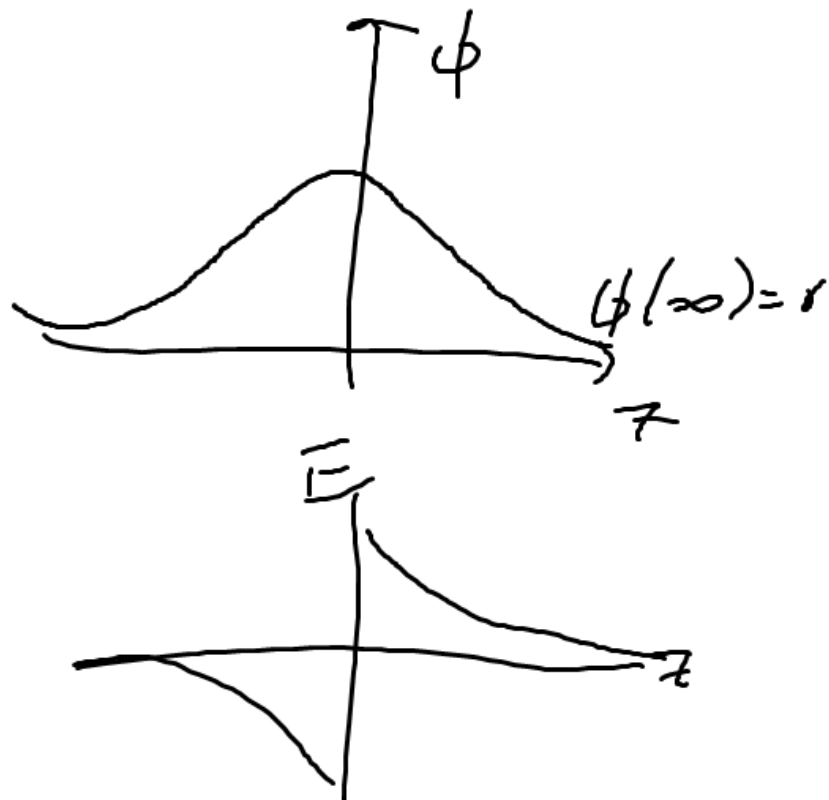


Figure 2.1

This can lead to a generalized conclusion about boundary conditions:

Example. Consider a box specifically Gauss's Box:

$$\oint \vec{E} \cdot \hat{n} dS = \sigma \frac{S}{\epsilon}$$

If we look at the coordinates perpendicular to the box which we can denote as

$$(E_{\perp}^{above} - E_{\perp}^{below}) S = \sigma \frac{S}{\epsilon_0}$$

$$(E_{\perp}^{above} - E_{\perp}^{below}) = \frac{\sigma}{\epsilon_0}$$

This is discontinuous and causes the electric field to jump for the perpendicular component. Now we can look at the parallel component with the surface. If we perform the loop integral over the section is that the electric field

$$\oint \vec{E} \cdot d\vec{l} = 0$$

which comes from the fact that $\nabla \times E = 0$. Thus we can write that

$$E_{||}^{above} - E_{||}^{below} = 0$$

$$\vec{E}^{above} - \vec{E}^{below} = \frac{\sigma}{\epsilon_0} \hat{n}$$

The potential across the boundary is going to be

$$\phi^{above} - \phi^{below} = \int_a^b \vec{E} \cdot d\vec{l} = 0$$

This tells us that the potential across a charged boundary is still continuous.

Note that we can convert quantities between ϕ , ρ , and E

$$\phi = - \int \vec{E} \cdot d\vec{l}$$

$$\nabla^2 \phi = - \frac{\rho}{\epsilon_0}$$

$$\nabla \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{E} = -\nabla \phi$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|^2} d\vec{r}'$$

$$\phi = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}'$$

Remark. Electrostatic Energy

The scalar potential is related to the work and energy that we care about. Suppose we wish to move a particle in an electric field from $p_0 \rightarrow p_1$ where the particle has charge q . What is the work?

$$W = - \int F \cdot d\vec{r}$$

$$W = - \int q\vec{E} \cdot d\vec{r}$$

$$= q [\phi(p) - \phi(p_0)]$$

Thus we see that the work is independent of the path and that the field $\vec{E}$ is conservative. If we know the work we can tell the difference to be

$$\phi(p) - \phi(p_0) = \frac{W}{q}$$

From here we can also know the work needed to assemble a particular charge distribution which is the work needed to assemble a distribution of charges q_1, q_2, q_3 . We know that $\phi(\infty) = 0$, so we can move the charges one by one. q_1 has no work done on it to move to the correct place. We then move each particle one by one. Thus for q_2 , we have

$$W_2 = - \int_{\infty}^{r_{12}} q_2 E_1 d\vec{r} = q_2 [\phi(\vec{r}_{12}) - \phi(\infty)] = q_1 \frac{q_2}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|}$$

We can also move the next particle which can be denoted as

$$W_3 = q_3 [\phi(\vec{r}_{13}) - 0] + q_3 [\phi(\vec{r}_{23}) - 0] = (q_1 + q_2) \frac{q_3}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|}$$

$$W_{total} = W_3 + W_2 + W_1$$

we see a structure to calculate the work to be defined to be the work done by 1 to 2, 1 to 3, and 2 to 3, denoted as

$$W_{total} = W_{12} + W_{13} + W_{23}$$

where

$$\begin{aligned} W_{ij} &= q_i \frac{q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|} \\ W_{total} &= \sum_{i,j,i \neq j, i < j} W_{ij} \\ &= \frac{1}{8\pi\epsilon_0} \sum_i q_i \sum_j \frac{q_j}{|\vec{r}_i - \vec{r}_j|} \\ &= \frac{1}{2} \sum_i q_i \phi_j(\vec{r}_i) \end{aligned}$$

For a continuous charge distribution we can rewrite this as

$$\begin{aligned} W &= \frac{1}{8\pi\epsilon_0} \int_V \int_{V'} \frac{\rho(\vec{r})\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV dV' \\ W &= \frac{1}{2} \int_V \rho(\vec{r}) \phi(\vec{r}) dV \end{aligned}$$

The energy can then be expressed in terms of work against the Coloumb interaction. Where in another way we can write

$$W = \frac{1}{2} \int_V \rho(\vec{r}) \phi(\vec{r}) dV$$

where the charge density is $\epsilon_0(\nabla \cdot \vec{E})$

$$= \frac{1}{2} \int_V \epsilon_0(\nabla \cdot \vec{E}) \phi dV$$

Notice that

$$\nabla \cdot (\vec{E}\phi) = \nabla\phi \cdot \vec{E} + (\nabla \cdot \vec{E})\phi = -|E|^2 + (\nabla \cdot \vec{E})\phi$$

We can now plug this back and rewrite the work as

$$W = \frac{1}{2} \int_V \epsilon_0 [\vec{\nabla} \cdot (\vec{E}\phi) + |E|^2] dV$$

Suppose that the charge is concentrated at a specific point in space. Then we integrate over the entire charge region. We have

$$\frac{\epsilon_0}{2} \left[\int_S \phi \vec{E} \cdot \hat{n} dS + \int_V |\vec{E}|^2 dV \right]$$

because there are no other charges, we can extend the first term of the surface to be infinitely large.

$$S \rightarrow r^2, |\vec{E}| \rightarrow \frac{1}{r^2}, \phi \rightarrow \frac{1}{r}$$

Thus the first term is proportional to $\frac{1}{r^3} r^2 \rightarrow 0$. Thus we can express the term simply as

$$W = \frac{\epsilon_0}{2} \int_V |\vec{E}|^2 dV$$

We can then denote W_e as the energy density associated with the electric field. And this is the other form for expressing the potential energy by calculating the electric field squared. Recall the form we had for the first form and there are some inconsistencies. For the charge density and potential, we can have negative charge density and a positive potential which means the total energy could be negative. On the other term, the term can never be negative. The way to reconcile this is integrate over a very small space. If we calculate the work to assemble the point charges, the first equation will blow up.

Lecture 6

2.4 Conductors

Feb 10 10:10

Definition 2.4.1. The conduction band is the band of electron orbitals where the valence band can jump to when excited. This allows the electrons to move freely in the material.

Definition 2.4.2. The valence band is the highest energy band normally occupied at zero temperature. In semiconductors and insulators, the valence band is the outermost band.

Definition 2.4.3. The band gap is the energy difference between the highest energy level of the valence band and the lowest energy level of the conduction band is called the band gap.

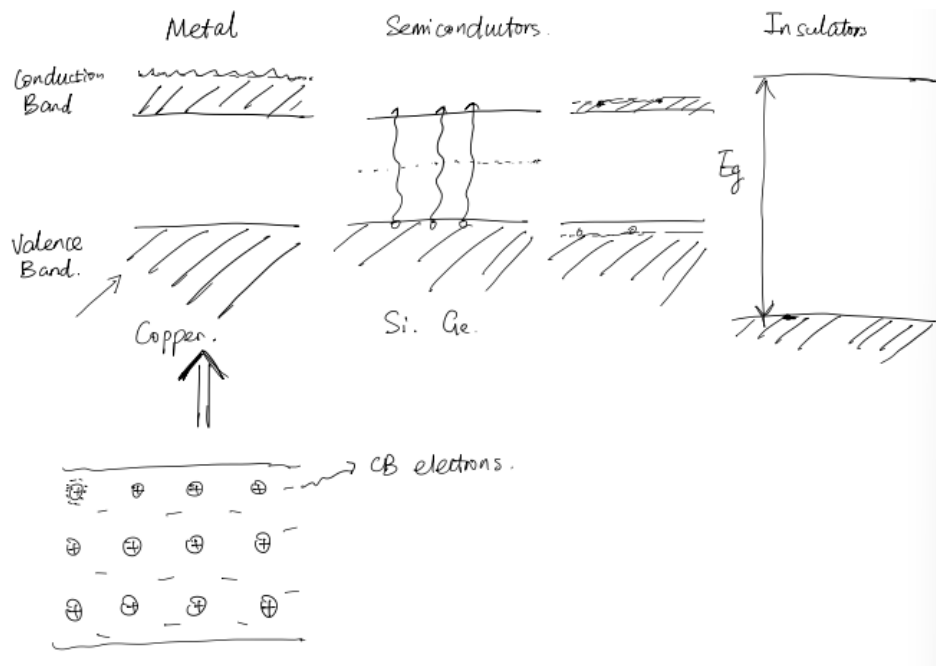


Figure 2.2

2.5 Ideal Conductors in Electrostatics

1. $\vec{E}$ inside a conductor is 0
2. $\nabla \vec{E} = \frac{\rho}{\epsilon_0} = 0$ inside the conductor (not the boundary)
3. Any net charge must reside on the surface (cannot be in the bulk)
4. Φ must be constant throughout the conductor
5. Electric field right outside the conductor is perpendicular s.t $\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}$

Remark. The solution to minimize the potential energy for a uniform volume distribution would be

$$W_{vol} = \frac{3Q^2}{20\pi\epsilon_0} \frac{1}{R}$$

for a surface distribution would be

$$W_{surf} = \frac{Q^2}{8\pi\epsilon_0} \frac{1}{R}$$

2.6 Induced Charges

Suppose we had a charge of positive q outside the surface of a conductor, we thus have an induced charge where the negatively charged electrons are pulled closer while the positive charged particles are moved away. For such a Electric field, it is still going to be 0 as the movement of the particles in the conductor are in such a way that the electric field is conserved. This is also known as a Faraday Cage.

Consider a charge in the cavity of the conductor, we must have that the induced charge of the positively charged cavity attracting negative charged particles and positive particles on the outside of the conductor.

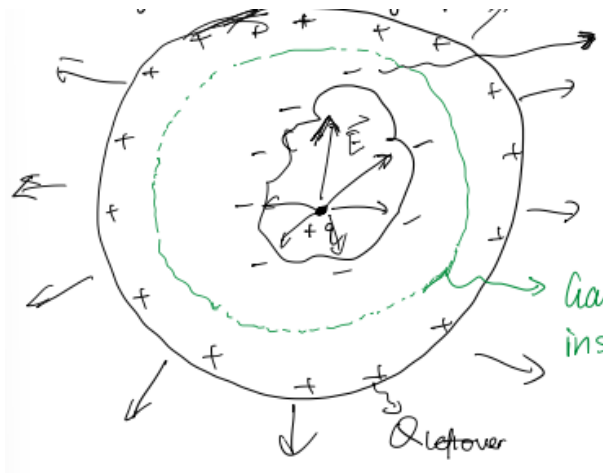


Figure 2.3

For Gauss's law we have that

$$\oint \vec{E} \cdot d\vec{s} = 0 = \frac{Q_{tot}}{\epsilon_0}$$

$$Q_{left} + Q_{induced} = 0 \Rightarrow Q_{left} = q$$

$$\vec{E}_{out} = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2}, \quad (r > R)$$

2.7 Force on the surface charge

$$\vec{F} = q\vec{E}$$

If we want to look at the force per unit area surface charge we must have that

$$\vec{f} = \sigma\vec{E}$$

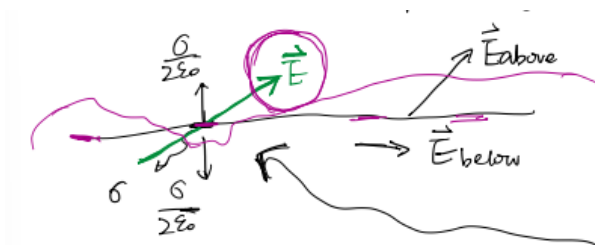


Figure 2.4

For the boundary conditions we must have that

$$\vec{E}_a - \vec{E}_b = \frac{\sigma}{\epsilon_0} \hat{n}$$

$$\Rightarrow \vec{E}_a = \vec{E} + \frac{\sigma}{2\epsilon_0} \hat{n}$$

$$\Rightarrow \vec{E}_b = \vec{E} - \frac{\sigma}{2\epsilon_0} \hat{n}$$

$$\vec{E} = \frac{\vec{E}_a + \vec{E}_b}{2}$$

If the electric field in a conductor is 0, then we must have that

$$\vec{E}_a = \frac{\sigma}{2\epsilon_0} \hat{n}$$

$$\vec{f} = \frac{\sigma^2}{2\epsilon_0} \hat{n}$$

which is the electrostatic pressure. Note that f doesn't depend on the sign of σ and always faces outward.

2.8 Capacitance

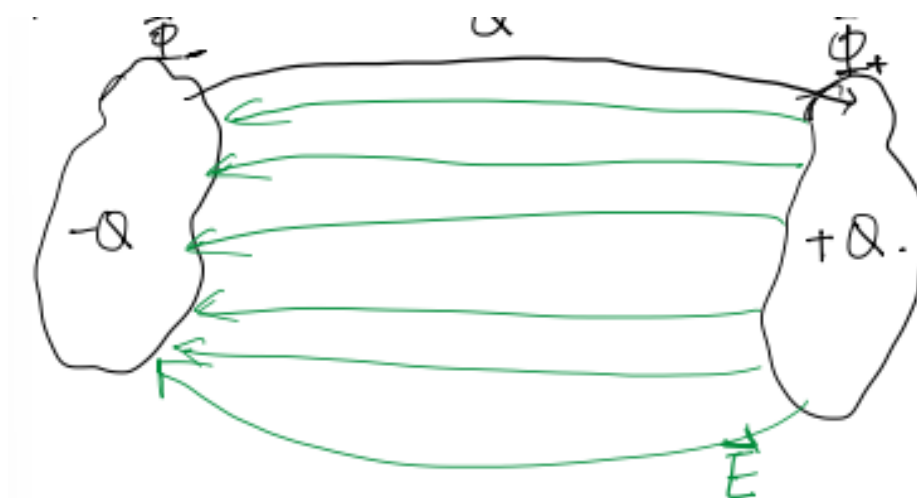


Figure 2.5

We know that the potential difference is given as

$$\Phi = \Phi_+ - \Phi_-$$

The capacitance is written as

$$C \equiv \frac{Q}{\Phi}$$

Recalling the definition of $\Phi(\vec{r})$ for volume charge density we see that the capacitance is a geometric quantity of the conductors i.e. (size, shape, separation). Now consider the relation to energy storage. The amount of work to move Q should be found as moving small dq

$$dW = dq \cdot \phi = dq \frac{q}{C}$$

$$W = \int dW = \int_0^c \frac{q}{C} dq = \frac{1}{2} C \Phi^2$$

Lecture 7

Example. One example to consider for conductors is a parallel plate capacitor. Consider two plates charged $+q, -q$, we must have that

$$\vec{E}_{out} = 0, \quad \vec{E}_{in} = \frac{\sigma}{\epsilon_0} \hat{n} = \frac{Q}{A\epsilon_0} \hat{z}$$

We can write the potential difference here as

$$\Phi = \int \vec{E} \cdot d\vec{l} = \frac{Q}{A\epsilon_0} d \equiv \frac{Q}{C}$$

$$C = \frac{A\epsilon_0}{d}$$

Feb 12 10:10

Theorem 2.8.1. Earnshaw's Theorem: It is possible to hold a charge in free space in a stable equilibrium by only using an electrostatic field.

1. No force on the charge $\Rightarrow \vec{E}(\vec{r}_0) = 0$
2. small displacement $\rightarrow$ force $\rightarrow$ pushback or a harmonic well

We must have a potential well where the work to displace it must be > 0

$$\Delta W = q\Delta\phi$$

$$\phi(\vec{r}) = \phi(\vec{r}_0) + \Delta\vec{r} \cdot \nabla \phi|_{\vec{r}=\vec{r}_0} + \frac{1}{2}(\Delta\vec{r} \cdot \vec{\nabla})^2 \phi|_{\vec{r}=\vec{r}_0} + \dots$$

For the first condition we have

$$E(\vec{r}) = 0 \Rightarrow \nabla\phi(\vec{r}_0) = 0$$

In x,y,z we have

$$\sum_{i=1}^3 \frac{1}{2}(\Delta x_i)^2 \frac{\partial^2 \phi}{\partial x_i^2} \Big|_{\vec{r}=\vec{r}_0}$$

For the second condition we have

$$\Delta\phi > 0$$

for all directions

$$\Delta = \Delta y = \Delta z$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi > 0$$

$$\nabla^2 \phi = -\frac{\rho}{\epsilon} > 0 \Rightarrow \rho(\vec{r}) \neq 0$$

If we assume that $\rho = 0$. At least one of the directions should have the derivative

$$\frac{\partial^2 \phi}{\partial j^2} < 0$$

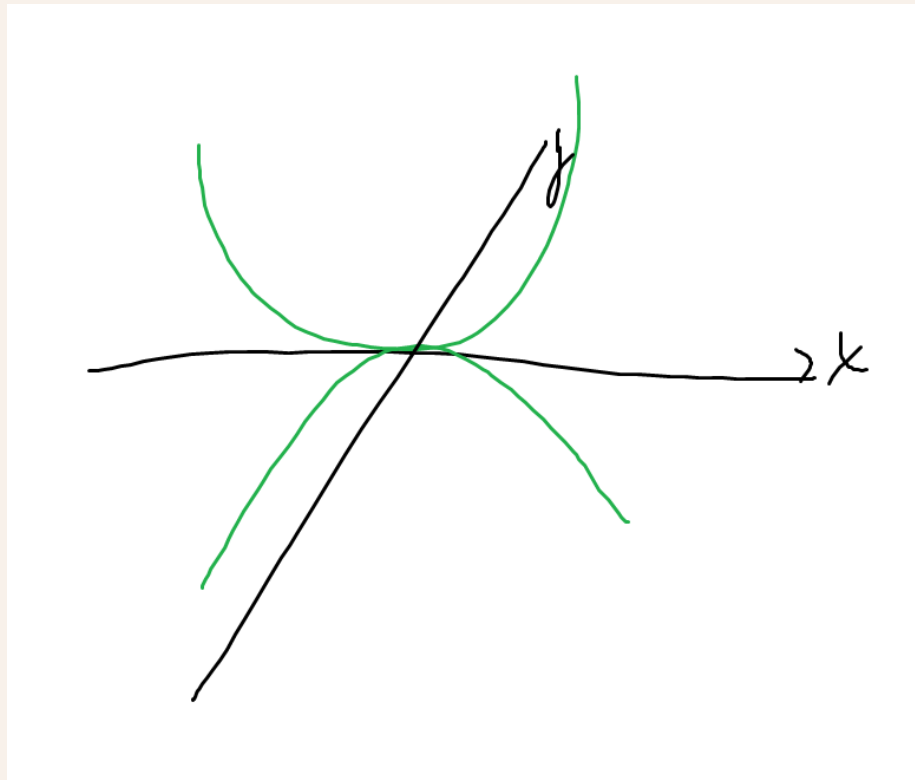


Figure 2.6

Trapped Ions

Example. This is useful for trapped ion quantum computing. There are two ways to hold a ion they use something called a Paul trap which uses a set of electrodes and alternates the electrodes which causes a rotating electric field or (AC field). This creates a quadruple moment which creates a trap in 3D.

The other method is called the **penning trap** where there is a static $\vec{E}$ field and a B field in the z direction.

Electrostatics for atoms, molecules, molecules, and crystals

Example. For the classical pictures of atoms (hydrogen atom), we have the following

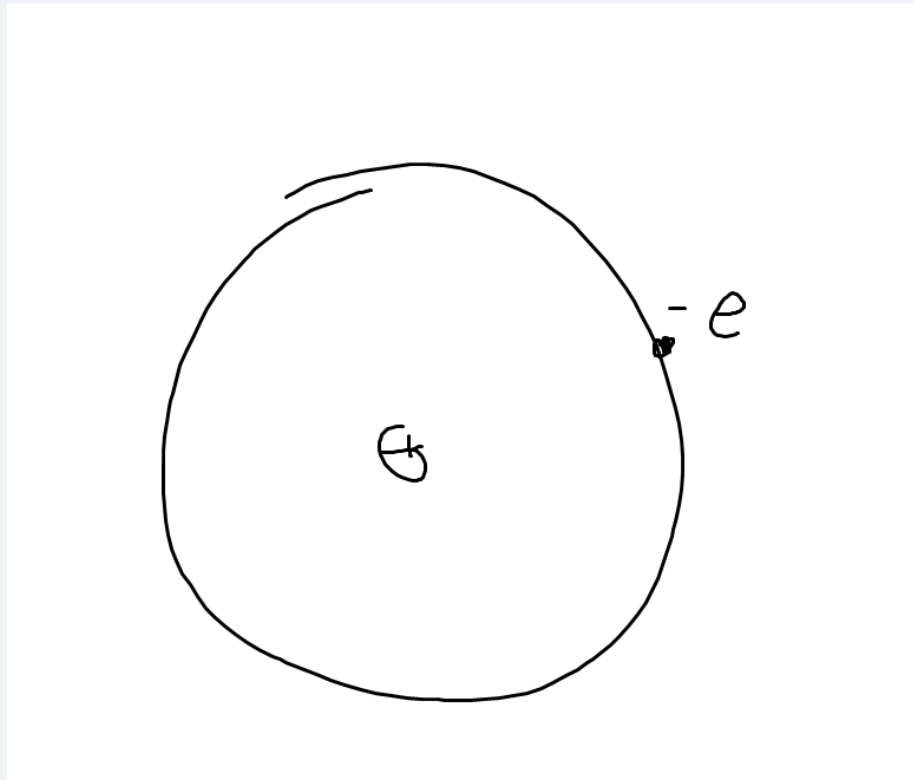


Figure 2.7

$$U = U_k + U_p$$

which is the total energy (kinetic and potential)

$$U_k = \frac{1}{2}mv^2$$

or defined in terms of angular momentum to be

$$\frac{L^2}{2mR^2}$$

The potential energy is going to be

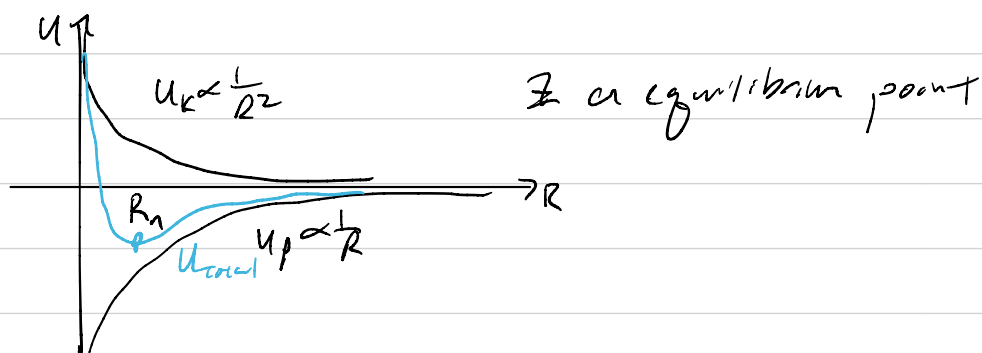
$$U_p = \frac{-q_e q_p}{4\pi\epsilon_0 R}$$

Additionally, Bohr Postulates that

$$L = n\hbar$$

Atomic Example:

For a given n :



we can take the derivative of the total energy:

$$\frac{dU}{dR} = -\frac{L^2}{mR^3} + \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = 0$$

$$= -\frac{1}{R} (2U_K + U_P) = 0 \Rightarrow U_K = \frac{1}{2} U_P$$

$$R_n = \frac{4\pi\epsilon_0}{me^2} (n\hbar)^2 \quad U_n = -\frac{me^4}{32(\pi\epsilon_0\hbar)^2}$$

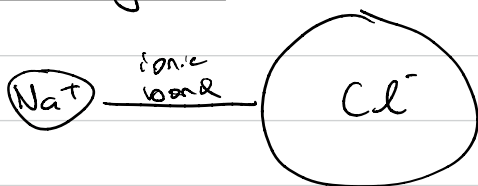
For $n=1$ ground state:

$$R_1 = 5.3 \times 10^{-11} \text{ m} \Rightarrow 0.53 \text{ \AA} = a_0 \text{ "Bohr's Radius"}$$

$$U_1 = -2.18 \times 10^{-18} \text{ J} \Rightarrow -13.6 \text{ eV (Ionization Energy)}$$



Ionic Crystals



face-centered cubic-structure:
like the fcc 3D structure



The lattice $d = 2.81 \text{ \AA}$

What would be the crystal binding energy "Energy required to separate the crystal into individual ions at infinite distance apart"

$2N \text{ ions} \sim N \text{ molecules}$

$$U_{\text{total}} = \frac{1}{2} \sum_{i=1}^{2N} q_i \phi_i = \frac{1}{8\pi\epsilon_0} \sum_{\substack{i,j \\ i \neq j}}^{2N} \frac{q_i q_j}{|\vec{r}_i - \vec{r}_j|}$$

Average Energy per Molecule

$$\frac{U_{\text{total}}}{N} \sim \frac{-e^2}{4\pi\epsilon_0} \sum_{j=2}^{2N} \frac{\pm 1}{|\vec{r}_i - \vec{r}_j|}$$

$$= -\frac{e^2}{4\pi\epsilon_0 R} \sum_{j=2}^{2N} \frac{\pm 1}{|\vec{r}_i - \vec{r}_j|/R}$$

"Madelung Constant" determined by the crystal structure.

For the crystal to be stable: $M > 0$

NaCl $M = 1.748$

$$\frac{U_{\text{total}}}{N} \approx -1.43 \times 10^{-18} \text{ J/molecule} = -8.95 \text{ eV/molecule}$$

Experimentally $\Rightarrow -8 \text{ eV/molecule}$

Chapter 3: Laplace's Equation

- Motivation

$$\rho \leftrightarrow \phi : \phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV'$$

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0} \text{ (Poisson equation)}$$

+ boundary condition. $\phi \rightarrow$ then we can easily solve for the potential

$(\phi, \frac{\partial \phi}{\partial \hat{n}} \text{ at boundary})$

If we have no free charges $\Rightarrow$ Poisson Eq $\rightarrow$ Laplace's Eq.

when the solutions are harmonic functions

$$\nabla^2 \phi = 0$$

Laplace. in 1-D

$$\pi: \frac{d^2 \phi}{dx^2} = 0$$

Chapter 3

Laplace

Lecture 8

3.1 Laplace's Equation

Feb 17 10:10

Laplace's equation is for the case where

$$\nabla^2 \Phi = 0$$

For Poisson we have a similar case where

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0} + B.C.$$

Consider Laplace's Equation in 1D for cartesian coordinates where

$$\frac{d^2 \Phi}{dx^2} = 0 \Rightarrow \Phi(x) = mx + b$$

The important conclusions we get is that

1. Any points within the region must have that $\Phi(x) = \frac{1}{2} (\Phi(x+d) + \Phi(x-d))$
2. The max and min must exist at the boundary and can be generalized to 2D and 3D case

Example. Spherical Coordinates

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 0$$

$$r^2 \frac{d\Phi}{dr} = C_1$$

$$\Phi = -\frac{C_1}{r} + C_2$$

which represents a point charge. For Cylindrical coordinate we have

$$\nabla^2 \Phi = \frac{1}{r} \frac{d}{dr} \left(r \frac{d\Phi}{dr} \right) = 0$$

$$\Phi = C_1 \ln r + C_2$$

which represents a line charge

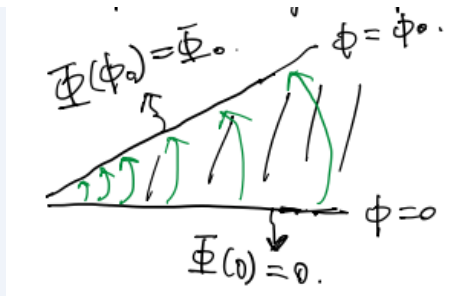


Figure 3.1

Example. For a wedged capacitor and given the conditions, we wish to find what $\Phi(\phi)$ is. We must have that

$$\frac{1}{r^2} \frac{d^2 \Phi}{d\phi^2} = 0$$

$$\Phi(\phi) = C_1 \phi + C_2$$

Plugging in initial conditions we have

$$\Phi(0) = 0 \Rightarrow C_2 = 0$$

$$\Phi(\phi_0) = C_1 \phi_0 = \Phi_0 \Rightarrow C_1 = \frac{\Phi_0}{\phi_0}$$

$$\Phi(\phi) = \frac{\Phi_0}{\phi_0} \phi$$

$$\vec{E} = \frac{1}{r} \frac{\Phi_0}{\phi_0} \hat{\phi}$$

In the 2D Laplace's Equation we have that

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0$$

1. $\Phi(x, y) = \frac{1}{2\pi R} \oint_{circle} \Phi dl$
2. No local max or min except for the boundaries

For the 3D case we have that

1. $\Phi(x, y, z) = \frac{1}{4\pi R^2} \oint_{sphere} \Phi ds$
2. No local max or min except boundaries $\Rightarrow$ Earnshaw's Theorem

$$\begin{aligned} \Phi_{avg} &= \frac{1}{4\pi r^2} \oint_s \Phi r^2 \sin \theta d\theta d\phi \\ \frac{d\Phi_{avg}}{dr} &= \frac{1}{4\pi} \oint_s \frac{d\Phi}{dr} \sin \theta d\theta d\phi \\ &= \frac{1}{4\pi r^2} \oint_s (\nabla \Phi \cdot \hat{r}) dS \\ &= \frac{1}{4\pi r^2} \int_V \nabla^2 \Phi dV = 0 \\ -\frac{1}{4\pi r^2} \int_S \vec{E} \cdot d\vec{s} &= -\frac{Q}{(4\pi r)^2 \epsilon_0} = 0 \end{aligned}$$

As $r \rightarrow 0$, $\Phi_{avg} = \Phi(x, y, z)$

Theorem 3.1.1. Uniqueness Theorem: We wish to show that the boundary conditions are unique.

Proof. Consider that we know $\rho(\vec{r})$ and Φ at the boundaries and there exists solutions Φ_1 and Φ_2 . We must have

$$\nabla^2 \Phi_1 = \frac{-\rho}{\epsilon_0} \quad \nabla^2 \Phi_2 = \frac{-\rho}{\epsilon_0}$$

$$\nabla^2 (\Phi_2 - \Phi_1) = 0 \quad \Phi_3 \equiv \Phi_2 - \Phi_1 \Rightarrow \nabla^2 \Phi_3 = 0$$

Thus we must have that $\Phi_3 = 0$ everywhere in the region since $\Phi_3|_{\text{boundaries}} = 0$. Thus, there only exists one solution.

Formally speaking, this is our 1st uniqueness theorem where we can say that the Poisson equation in some region is uniquely determined if the value of Φ is specified on all boundaries (charge density is specified)

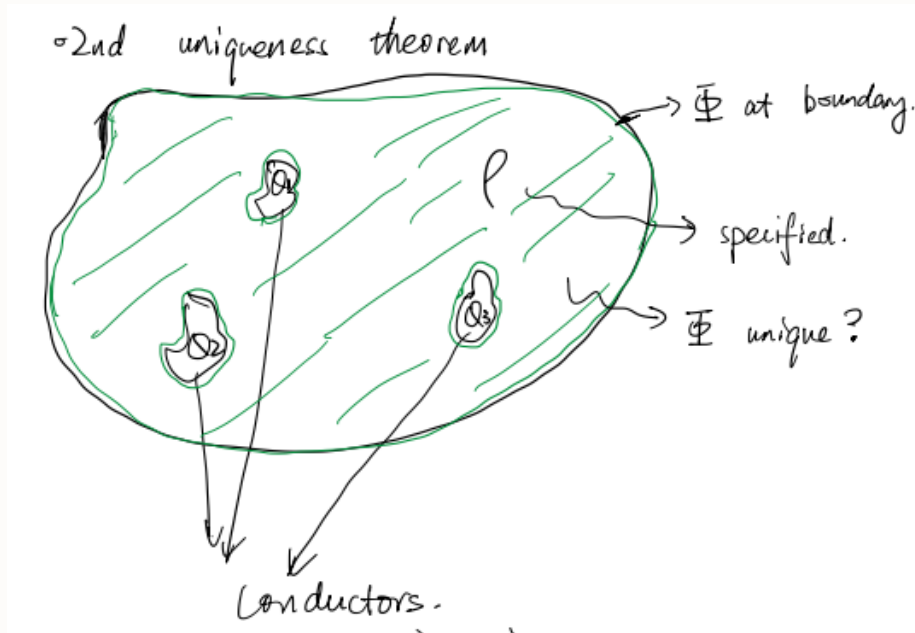


Figure 3.2

For the second uniqueness theorem, consider we know ρ , Φ at boundaries, and some charges $q_1, q_2, \dots$ inside acting as conductors. Assuming that we have two fields $\vec{E}_1, \vec{E}_2$ we must have

$$\nabla \vec{E}_1 = \frac{\rho}{\epsilon_0} \quad \nabla \vec{E}_2 = \frac{\rho}{\epsilon_0}$$

For each conductor we have the relationship

$$\oint \vec{E}_j \cdot d\vec{S} = \frac{Q_j}{\epsilon_0}$$

$$\vec{E}_3 \equiv \vec{E}_1 - \vec{E}_2 \Rightarrow \nabla \cdot \vec{E}_3 = 0 \Rightarrow \oint \vec{E}_3 \cdot d\vec{S} = 0$$

$$\nabla \cdot (\Phi_3 \vec{E}_3) = \Phi_3 (\nabla \cdot \vec{E}_3) + \vec{E}_3 \cdot (\nabla \Phi_3) = -|\vec{E}_3|^2$$

We then have

$$-\int |\vec{E}_3|^2 dV = \int \nabla \cdot (\Phi_3 \vec{E}_3) dV = \int_S \Phi_3 \vec{E}_3 \cdot d\vec{S} = \oint_S \Phi_3 \vec{E}_3 \cdot d\vec{S} + \sum_j \Phi_3 \oint_S \vec{E}_3 \cdot d\vec{S}$$

the two last terms must be 0 thus we must have that $\vec{E}_3 = 0$ everywhere in space. ■

Lecture 9

3.2 Method of Images

Feb 19 10:10

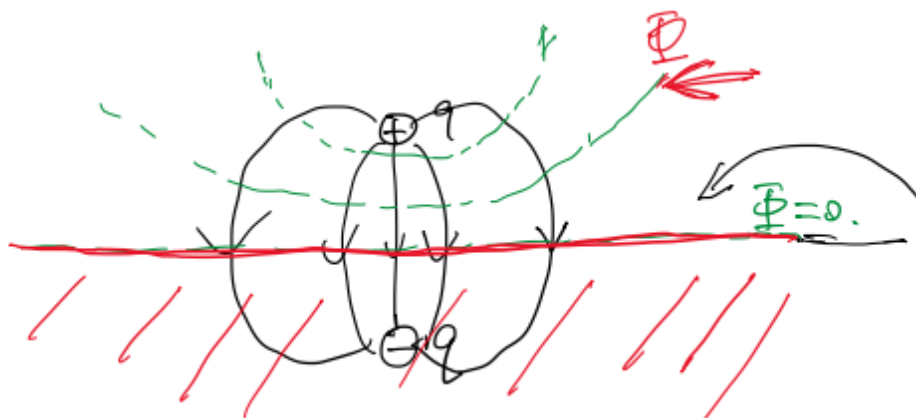


Figure 3.3

As shown in the figure, suppose we wanted to solve for a particle and a sheet. We can inact the method of images which helps us solve certain boundary conditions. Our potential can be written as

$$\Phi(\vec{r}) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} - \vec{r}_+|} - \frac{1}{|\vec{r} - \vec{r}_-|} \right]$$

Given the boundary we know that

1. $\Phi(z=0) = 0$
2. $\vec{E} \perp \text{surface}$
3. $\Phi \rightarrow 0, r \gg d$

From the boundary conditions we deduce

$$-\frac{\sigma}{\epsilon_0} = \frac{\partial\Phi_{above}}{\partial n} = -\epsilon_0 \left. \frac{\partial\Phi}{\partial z} \right|_{z=0}$$

since the surface is a conductor.

$$\begin{aligned} \sigma &= -\epsilon_0 \frac{\partial\Phi_{above}}{\partial n} = -\epsilon_0 \left. \frac{\partial\Phi}{\partial z} \right|_{z=0} \\ &= -\frac{q}{4\pi} \left[-\frac{(z-d)}{(x^2+y^2+(z-d)^2)^{\frac{3}{2}}} - \frac{(z+d)}{(x^2+y^2+(z+d)^2)^{\frac{3}{2}}} \right] \Big|_{z=0} \\ &= -\frac{qd}{2\pi [x^2+y^2+d^2]^{\frac{3}{2}}} \end{aligned}$$

The total induced charge is given by

$$\begin{aligned} Q &= \int \sigma dS \\ &= \frac{1}{2\pi} \int_0^\infty r dr \int_0^{2\pi} d\phi \cdot \frac{-qd}{(r^2+d^2)^{\frac{1}{2}}} \\ &= \int_0^\infty \frac{-rqddr}{(r^2+d^2)^{\frac{3}{2}}} = qd \left. \frac{1}{(r^2+d^2)^{\frac{1}{2}}} \right|_0^\infty = -q \end{aligned}$$

Example. Suppose we had two charges where $q' = -\gamma q$ ($0 < \gamma < 1$)

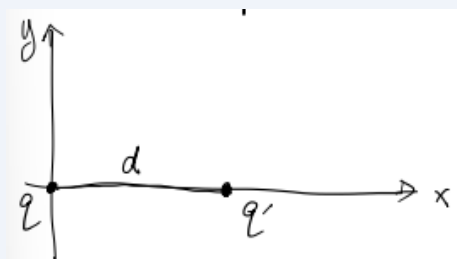


Figure 3.4

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{x^2 + y^2 + z^2}} + \frac{q'}{\sqrt{(x-d)^2 + y^2 + z^2}} \right) = \Phi_0$$

We have the condition that $\Phi_0 = 0$

$$\Rightarrow 0 = \frac{1}{\sqrt{x^2 + y^2 + z^2}} - \frac{\gamma}{\sqrt{(x-d)^2 + y^2 + z^2}}$$

Solving we must have the form

$$\left(x - \frac{d}{1-\gamma^2} \right)^2 + y^2 + z^2 = \left(\frac{d\gamma}{1-\gamma^2} \right)^2$$

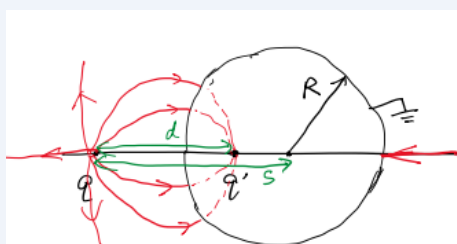


Figure 3.5

We simply have a sphere of radius $\frac{d\gamma}{1-\gamma^2}$. From the conditions of the sphere we must have that q' is inside the sphere while q is outside of the sphere. From these, conditions, we can write that

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{d^2} = -\frac{1}{4\pi\epsilon_0} \frac{q^2 \frac{R}{s}}{(s^2 - R^2)^2 / s^2} = -\frac{1}{4\pi\epsilon_0} \frac{q^2 R s}{(s^2 - R^2)^2}$$

where $s = \frac{d}{1-\gamma^2}$

3.3 Numerical Techniques for Laplace's Equation

Given either Poisson's or Laplace's, we can perform the following methods

1. Finite difference: simple
2. finite element (minimizing lagrangian): good for irregular geometry
3. Spectral methods (FFT): fast

Lecture 10

3.4 Separation of Variables

Feb 26 10:10

The PDE we have would be easier to solve if we converted it to ODEs. Thus, for Laplace's equation in Cartesian we have

$$\sum_{i=1}^3 \frac{\partial \Phi}{\partial x_i} = 0$$

Assuming that $\Phi(x, y, z) = X(x)Y(y)Z(z)$ we observe the separation of differential equations to be

$$YZ \frac{d^2 X}{dx^2} + XZ \frac{d^2 Y}{dy^2} + XY \frac{d^2 Z}{dz^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

We can write each part of the latter equation as

$$\begin{aligned} f_x(x) + f_y(y) + f_z(z) &= 0 \\ \Rightarrow f_x(X) = C_1, f_y(Y) = C_2, f_z(Z) = C_3 \\ C_1 + C_2 + C_3 &= 0 \end{aligned}$$

Note that in the 2D case we must have $C_1 = -C_2 = \alpha_m^2$

$$X(x) = e^{\alpha x}, \quad \frac{dX}{dx} = \alpha e^{\alpha x}$$

$$\frac{d^2 X}{dx^2} = \alpha^2 e^{\alpha x}, \quad \alpha \geq 0$$

If we let $\alpha = i\beta$ then we must have that $\ddot{X} = \beta^2 e^{i\beta x}$. Going back to the 3D case, we see that there are only two possibilities:

$$C_1, C_2 > 0, C_3 < 0, \quad C_1 > 0, C_2, C_3 < 0$$

Thus we must have the equation

$$\begin{aligned} \alpha^2 + \beta^2 &= \gamma^2 \\ C_1 = -\alpha^2, C_2 = -\beta^2, C_3 &= \gamma^2 \end{aligned}$$

The solutions are of the form of $e^{ik_i x_i}$ or e^{kx}

$$X(x) = A \cos(\alpha x) + B \sin(\alpha x)$$

$$Y(y) = C \cos(\beta y) + D \sin(\beta y)$$

$$Z(z) = E e^{\gamma z} + F e^{-\gamma z}$$

Therefore the solutions of the differential equations are of the form where we can sum up all solutions with their given coefficients as

$$\sum_{m=1}^{\infty} X_m(x), \quad \sum_{n=1}^{\infty} Y_n(y), \quad \sum_{m,n=1}^{\infty} Z_{mn}(z)$$

$$\gamma_{mn}^2 = \alpha_m^2 + \beta_n^2 > 0$$

$$\Phi(x, y, z) = X(x)Y(y)Z(z)$$

Example. Suppose we had a rectangular prism with $\Phi(y)$ with the given boundary conditions

1. $\Phi(x, 0) = \Phi(x, a) = 0$
2. $\Phi(0, y) = \Phi(y)$
3. $\lim_{x \rightarrow \infty} \Phi(x, y) \rightarrow 0$

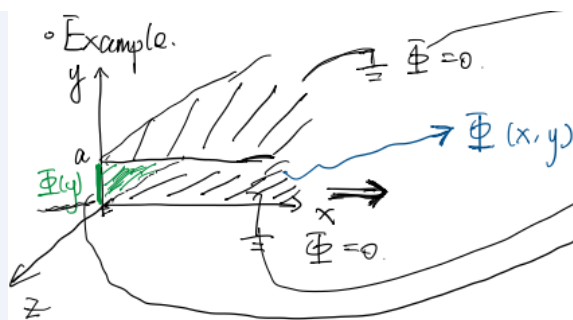


Figure 3.6

Given the boundary conditions $y \propto \cos$ and $x \propto \exp$. Thus we can write the solution in the form

$$\Phi(x, y) = \sum_{m=1}^{\infty} (A_m e^{\alpha_m x} + B_m e^{-\alpha_m x}) (C_m \cos(2my) + D_m \sin(2my))$$

Applying the boundary conditions we must have

$$\Phi(x, 0) = 0 \Rightarrow C_m = 0, A_m = 0$$

$$\Phi(x, y) = \sum_{m=1}^{\infty} B_m e^{-\alpha_m x} D_m \sin(\alpha_m y)$$

Let $G_m \equiv B_m D_m$ and applying the additional conditions that

$$\Phi(x, a) = 0 \Rightarrow \sin(\alpha_m a) = 0 \Rightarrow \alpha_m = \frac{m\pi}{a}$$

$$\Phi(0, y) = \Phi(y) \Rightarrow \sum_{m=1}^{\infty} G_m \sin(\alpha_m y)$$

To solve for the coefficient of G_m we simply have to normalize and solve the case where

$$\sum_{m=1}^{\infty} G_m \int_0^a \sin(\alpha_m y) \sin(\alpha_n y) dy = \int_0^a \Phi(y) \sin(\alpha_n y) dy$$

$$\int_0^a \sin(\alpha_m y) \sin(\alpha_n y) dy = \frac{n\pi}{2\alpha_n} \delta_{nm} = \frac{a}{2} \delta_{nm}$$

Solving for G_n we find if $\Phi(y) = \Phi_0$

$$G_n = \begin{cases} 0, n \text{ even;} \\ \frac{4\Phi_0}{n\pi}, n \text{ odd;} \end{cases}$$

$$\Phi(x, y) = \frac{4\Phi_0}{\pi} \sum_{m=1}^{\infty} \frac{1}{m} e^{-\frac{m\pi}{a} x} \sin\left(\frac{m\pi y}{a}\right)$$

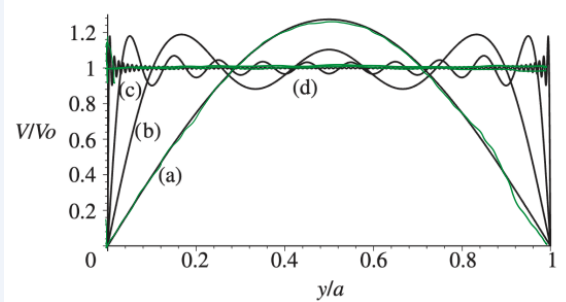


Figure 3.7

Example. Consider another example where instead we have a insulator defined by the boundary conditions

1. $\Phi(x, 0, z) = \Phi(x, a, z) = \Phi(x, y, 0) = \Phi(x, y, a) = 0$
2. $\lim_{x \rightarrow \infty} \Phi(x, y, z) \rightarrow 0$
3. $\frac{\partial \Phi}{\partial x} \Big|_{x=0} = -\frac{\sigma(y, z)}{\epsilon_0}$

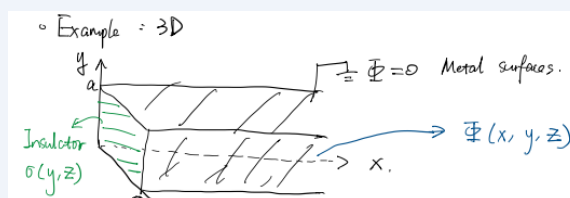


Figure 3.8

Given the boundary conditions. x must decay from the plate. We must have sin functions for the y, z coordinates. Thus, dropping indexes we must have that

$$\Phi(x, y, z) = \sum_{m,n=1}^{\infty} B e^{-\alpha x} D_n \sin(\beta y) F_m \sin(\gamma z)$$

$$\alpha^2 = \beta^2 + \gamma^2, \alpha > 0$$

Let $G = BDF$ then solving the boundary conditions we have

$$\beta = \frac{n\pi}{a}, \quad \gamma = \frac{m\pi}{a}, \quad \alpha = \sqrt{m^2 + n^2} \frac{\pi}{a}$$

$$\frac{\partial \Phi}{\partial x} \Big|_{x=0} = \frac{-\sigma(y, z)}{\epsilon_0} = \sum_{m,n=1}^{\infty} -\alpha G \sin(\beta y) \sin(\gamma z)$$

$$\frac{1}{\epsilon_0} \int_0^a \int_0^a \sigma \sin(\beta y) \sin(\gamma z) dy dz$$

$$= \sum_{m,n} \alpha G \int_0^a \sin(\beta y) \sin(\beta y) dy \int_0^a \sin(\gamma z) \sin(\gamma z) dz$$

Reintroducing indexes notice the kronecker delta in each of the integrals. Thus calculating G_{KL} we simply multiply by itself s.t. the integrals disappear

$$G_{KL} = \frac{1}{\epsilon_0 \alpha} \left(\frac{2}{a} \right)^2 \int_0^a \int_0^a \sigma(y, z) \sin(\beta_K y) \sin(\gamma_L z) dy dz$$

Lecture 11

3.5 Spherical Coordinates

Mar 3 10:10

Recall Laplace's equation in 3D

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$

$$\Phi(\vec{r}) = R(r)\Theta(\theta)Q(\phi), \quad 0 < \theta < \pi, 0 < \phi < 2\pi$$

Since the form of the solution is separable, we can solve in component form

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\frac{1}{Q} \frac{d^2 Q}{d\phi^2}$$

The equation must equal to a constant since they are all differentiated w.r.t different terms. Thus, denote $-m^2$ as the constant

$$\frac{d^2 Q}{d\phi^2} = -m^2 Q, \quad Q(\phi) = Q(\phi + 2n\pi)$$

$$Q(\phi) \propto e^{im\phi} \quad (m = 0, \pm 1, \pm 2, \dots)$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{m^2}{\sin^2 \theta} = \text{const}$$

Make the following substitution where

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = L(L+1)R$$

$$R(r) = Ar^L + \frac{B}{r^{L+1}}$$

Let $\theta \Rightarrow x = \cos \theta$ then we have

$$\frac{d}{dx} \left[(1-x^2) \frac{d\Theta}{dx} \right] + \left[L(L+1) - \frac{m^2}{1-x^2} \right] \Theta = 0$$

For $m = 0$, there exists azimuthal symmetry where

$$(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + L(L+1)\Theta = 0$$

which is known as Legendre's equation. The legendre polynomial associated with the equation is given as

$$\Theta(\theta) = P_L(x) = P_L(\cos \theta)$$

$$\frac{1}{\sqrt{1-2xw+w^2}} = \sum_{L=0}^{\infty} P_L(x)w^L$$

$$P_L(x) = \frac{1}{2^L L!} \left(\frac{d}{dx} \right)^L (x^2-1)^L \quad (\text{Rodrigues formula})$$

From here we note the orthogonality relationship

$$\int_{-1}^1 P_L(x) P_K(x) dx = \frac{2}{2L+1} \delta_{KL}$$

From here we can find $\Phi(r, \theta, \phi)$ which is given as

$$\Phi = \left(A_L r^L + \frac{B}{r^{L+1}} \right) P_L \cos(\theta)$$

For a general solution where $m = 0$

$$\Phi(r, \theta, \phi) = \sum_{L=0}^{\infty} \left(A_L r^L + \frac{B_L}{r^{L+1}} \right) P_L \cos \theta$$

For solution $m \neq 0$

$$\Phi(r, \theta, \phi) = \sum_{L=0}^{\infty} \sum_{m=-L}^L \left(A_L r^L + \frac{B_L}{r^{L+1}} \right) \sqrt{\frac{2L+1}{4\pi} \frac{(L-m)!}{(L+m)!}} P_L^M(\cos \theta) e^{im\phi}$$

We denote the last term with the square root and what follows as the spherical harmonics for the H atom also known as $Y_L^M(\theta, \phi)$

Example. Consider an example of a charged sphere inside a uniform E field where $\vec{E}_0 = E_0 \hat{z}$

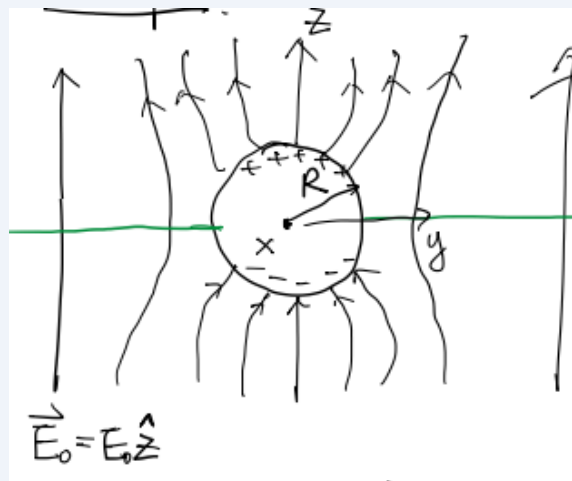


Figure 3.9

$$\vec{E} = \vec{E}_0 + \vec{E}_{ind}$$

$$\Phi = \phi_0 + \phi_{ind}$$

$$\Phi_0 = - \int_{r_0}^r \vec{E}_0 \cdot d\vec{l} = -E_0 \int_{z_0}^z dz = -E_0(z - z_0)$$

Choose $z_0 = 0, \phi_0 = 0$ we have

$$\Phi_0 = -E_0 z = -E_0 r \cos \theta = -E_0 r P_1(\cos \theta)$$

Applying our boundary conditions

1. $\Phi = 0$ when $r \leq R$
2. $\Phi_{r \rightarrow \infty} \rightarrow \Phi_0 = -E_0 r P_1(\cos \theta)$

For azimuthal symmetry ($m = 0$)

$$\Phi(r, \theta, \phi) = \sum_{L=0}^{\infty} \left(A_L r^L + \frac{B_L}{r^{L+1}} \right) P_L(\cos \theta)$$

Applying the B.C.(2)

$$r \rightarrow \infty \Rightarrow -E_0 r P_1(\cos \theta)$$

$$L = 1, \quad A_1 = -E_0$$

$$L \neq 1, \quad A_L = 0$$

Apply B.C.(1) $\Phi(R, \theta, \phi) = 0$

$$-E_0 R P_1 \cos \theta + \sum_{L=1}^{\infty} \frac{B_L}{R^{L+1}} P_L \cos \theta = 0$$

From here we have the orthogonality relationship that

$$E_0 R \int_0^\pi P_1 \cos \theta P_K \cos \theta \sin \theta d\theta = \sum_{L=0}^{\infty} \frac{B_L}{R^{L+1}} \int_0^\pi P_L \cos \theta P_K \cos \theta \sin \theta d\theta$$

We know that the integrals are simply given as the Kronecker deltas thus we have that

$$\frac{2}{3} E_0 R \delta_{LK} = \frac{2}{2K+1} \frac{B_L}{R^{L+1}} \delta_{LK}$$

$$B_1 = E_0 R^3, B_L = 0 (L \neq 1)$$

$$\Phi(r, \theta, \phi) = -E_0 r \cos \theta + E_0 \frac{R^3}{r^2} \cos \theta$$

The first term is the constant field $\vec{E}$ and the second term is from the induced charges we get. From here we can write our electric field as the negative gradient of our potential which gives

$$E_r = E_0 \left(1 + 2 \left(\frac{R}{r} \right)^3 \right) \cos \theta$$

$$E_\theta = -E_0 \left[1 - \left(\frac{R}{r} \right)^3 \right] \sin \theta$$

$$\sigma(\theta) = \epsilon_0 E_r(R, \theta, \phi) = 3\epsilon_0 E_0 \sin \theta$$

3.6 Cylindrical Coordinates

We have Laplace's equation

$$\nabla^2 \Phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0$$

Assume the solution

$$\Phi(r, \phi, z) = R(r)Q(\phi)Z(z)$$

$$\frac{d^2 Q}{d\phi^2} + n^2 Q = 0 \Rightarrow Q(\phi) \propto e^{in\phi}, \quad n = 0, \pm 1, \pm 2$$

$$\frac{d^2 Z}{dz^2} - k^2 z = 0 \Rightarrow Z(z) = \propto e^{\pm kz}$$

$$r \frac{d}{dr} \left(r \frac{dR}{dr} \right) + (k^2 r^2 - n^2) R = 0$$

Let $u = kr$

$$u^2 \frac{d^2 R}{du^2} + u \frac{dR}{du} + (u^2 - n^2) R = 0$$

This is known as Bessel's equation. The solution of this is given as

$$\Phi(r, \phi, z) = \sum_{m,n=1}^{\infty} [A_{mn} J_n(k_m r) + B_{mn} N_n(k_m r)] e^{\pm in\phi \pm k_m z}$$

Lecture 12

3.7 Review

Mar 5 10:10

Separation of variables with spherical coordinates will give a special function specifically the spherical harmonics

$$Y_v^m(\theta, \phi) e^{im\phi} P_i^m \cos \theta$$

For the cylindrical coordinates we have the Bessel function

$$e^{in\phi}, e^{\pm kz}, \text{ Bessel function}$$

3.8 Cylindrical Coordinates

$$\Phi(r, \phi, z) \approx \sum_{m,n} [A_{mn} I_n(k_m r) + B_{mn} N_n(l_m r)] e^{\pm in\phi \pm kmz}$$

Example. We have a charged wire where ϕ_0 is the flux inside and ϕ on the surface and $\phi = 0$ for above the surface. We know from here that no charge $\Rightarrow B_{mn} = 0$; symmetry $\Rightarrow n = 0$. Thus we get the equation

$$\Phi(r, \phi, z) = \sum_m A_m J_0(k_m r) e^{-k_m z}$$

From our boundary conditions we have $\Phi(a, \phi, z) = 0$ This type of boundary condition tells the roots and our harmonic is the bessel function. Another boundary condition is

$$\Phi(r, \phi, 0) = \phi_0 \rightarrow A_m$$

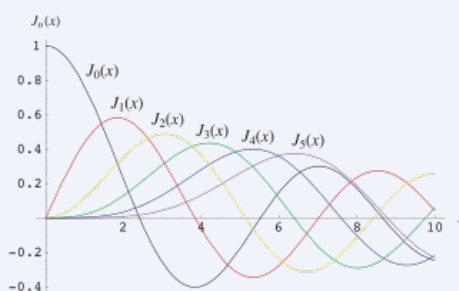


Figure 3.10

The orthogonality condition we get here is simply

$$\int_0^a J_n(k_m r) J_n(k_L r) dr = \frac{a^2}{2} J_{n+1}^2(k_L a) \delta_{mL}$$

3.9 Multipole Expansion

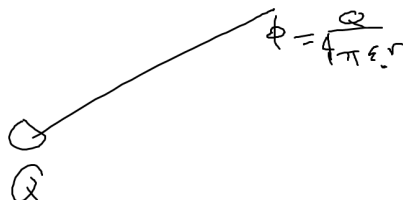


Figure 3.11

figure Suppose we had a dipole here, then we will get the form that the potential to be

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

$$\frac{1}{r_{\pm}} = \frac{1}{r \sqrt{1 \mp \frac{d}{r} \cos \theta}} \rightarrow \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

$$\frac{1}{r_+} - \frac{1}{r_-} = \frac{d}{r^2} \cos \theta$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2} \propto \frac{1}{r^2}$$

Thus for monopoles the potential falls off as a function of $\frac{1}{r}$ and the dipole will fall off as a function of $\frac{1}{r^2}$. For quadrupoles we get that the potential falls as a function as $\frac{1}{r^3}$. If we continue with a cube we will have that the potential fall off as a function of $\frac{1}{r^4}$. Thus if we are interested in a far-field approximation we only care about the monopole term, and if 0 then consider the dipole term and continue. This is similar to when expanding the potential term as a function of $\frac{1}{r}$. Thus we can do the following expansion for general charge distributions:

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV'$$

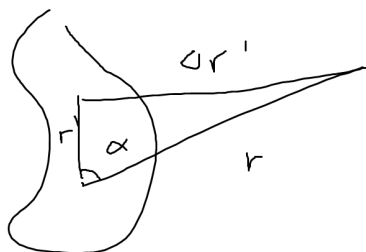


Figure 3.12

Suppose that we want to calculate the potential $\vec{r}$ away we get that

$$(\Delta r')^2 = r^2 + r'^2 - 2rr' \cos \alpha = r^2 \left[1 + \left(\frac{r'}{r} \right)^2 - 2 \frac{r'}{r} \cos \alpha \right]$$

$$\Delta r' = r\sqrt{1 + \epsilon}, \epsilon = \frac{r'}{r} \left(\frac{r'}{r} - 2 \cos \alpha \right)$$

For $\frac{r'}{r} \rightarrow 0, \epsilon \rightarrow 0$

$$\begin{aligned} \frac{1}{\Delta r'} &= \frac{1}{r} (1 + \epsilon)^{-\frac{1}{2}} = \frac{1}{r} \left(1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \dots \right) \\ &= \frac{1}{r} \left[1 + \frac{r'}{r} + \left(\frac{r'}{r} \right)^2 \left(\frac{3 \cos^2 \alpha - 1}{2} \right) + \dots \right] \end{aligned}$$

which are the Legendre polynomials.

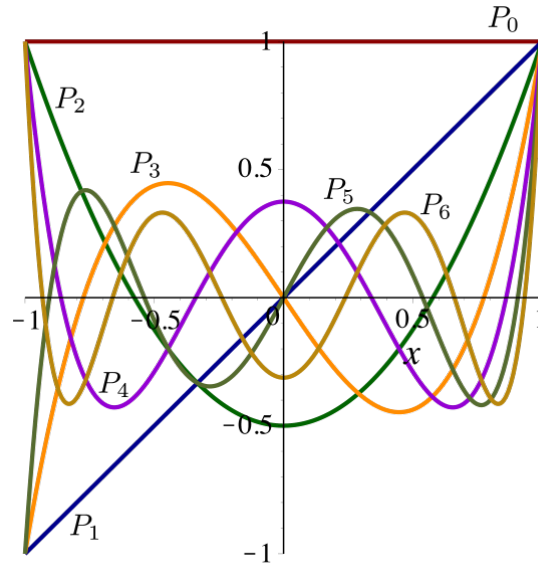


Figure 3.13

$$\frac{1}{\Delta r'} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \alpha)$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \int \rho(\vec{r}') dV' + \frac{1}{r^2} \int r' \cos \alpha \rho(\vec{r}') dV' + \dots \right]$$

We see that the first term gives the monopole and the second term gives the dipole and the third term will give you the quadrupole. Notice that for the physical dipole we are already making some assumptions. However, the formula above for $\Phi(\vec{r})$ is an exact equation when taking account of higher order terms. Notice the dipole term

$$\begin{aligned} \Phi_{dip}(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int r' \cos \alpha \rho(\vec{r}') dV' \quad r' \cos \alpha = \hat{r} \cdot \vec{r}' \\ &= \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r} \cdot \int r' \rho(\vec{r}') dV' \\ \Phi_{dip}(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r} \cdot \vec{p} \end{aligned}$$

where the geometry of the charge distribution is

$$\vec{p} = \sum_i q_i \vec{r}'_i = q \vec{d}$$

Remark. For the physical dipole, it gives the dipole term only when $\frac{d}{r} \rightarrow 0$. Thus, to construct a mathematical dipole we would need d to be very small. We see that for the mathematical dipole we have $\vec{p} = q\vec{d}$. Thus we must have the condition that $d \rightarrow 0$ and $q \rightarrow \infty$ in order to make $\vec{p}$ finite.

3.10 Dependence on Coordinate Origin

Suppose we placed a charge a distance d on the y -axis from the origin.

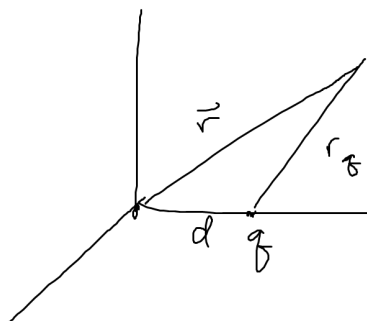


Figure 3.14

$$\Phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r_q}$$

We see that this has a net dipole moment of $\vec{p} = qd\hat{y}$ which would have a nonzero dipole moment. Now suppose we wanted to calculate the dipole in shifted coordinates where

$$\begin{aligned} \tilde{\vec{p}} &= \int \tilde{\vec{r}}' \rho(\vec{r}') dV' = \int (\vec{r}' - \vec{a}) \rho(\vec{r}') dV' \\ &= \int \vec{r}' \rho(\vec{r}') dV' - \vec{a} \int \rho(\vec{r}') dV' \\ &= \vec{p} - \vec{a}Q \end{aligned}$$

Thus, we ask where the origin is based on the dipole.

3.11 Electric field of Dipole

$$\Phi_{dip} = \frac{\hat{r} \cdot \vec{p}}{4\pi\epsilon_0 r^2} = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$$

where the dipole is aligned with the z -axis. We have

$$E_r = -\frac{\partial \Phi_{dip}}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial \Phi_{dip}}{\partial \theta}, \quad E_\phi = -\frac{1}{r \sin \theta} \frac{\partial \Phi_{dip}}{\partial \phi} = 0$$

$$\vec{E}_{dip} = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

If we plot this given that we have an ideal small dipole we have

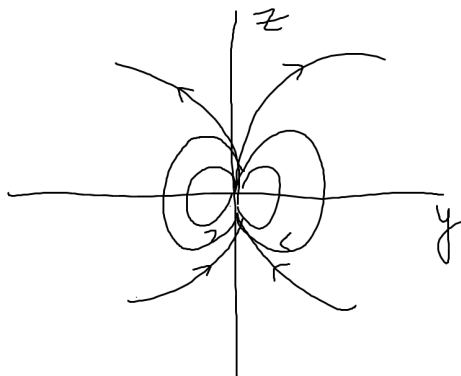


Figure 3.15

If the total charge is nonzero we must find the dipole. If there is something delicate, we can use the quadrupole terms and more which tell us how the field behaves in far-field.

Chapter 4

Electric Fields in Matter

4.1 Polarization

Recall that conductors have positively charged ions fixed to the lattice points while the electrons are freely moving. We can also consider dielectrics. Dielectrics are materials that don't allow current to flow. They are more often called insulators because they are the exact opposite of conductors.

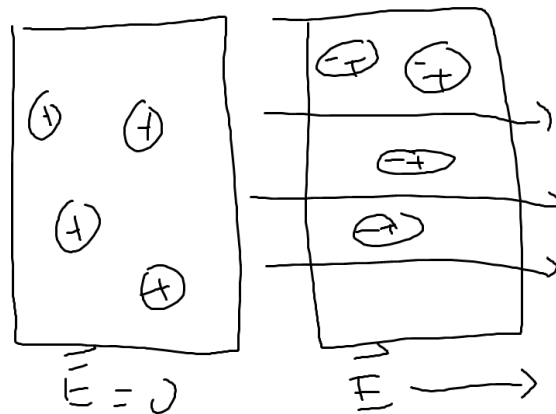


Figure 4.1

This gives rise to a macroscopic polarization that we get to see. For an induced dipole which we see here we get a net dipole moment or a simplified model. We can show that this can be represented in the following:

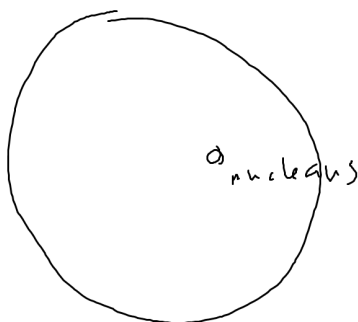


Figure 4.2

When we apply a $\vec{E} \rightarrow$ to the right, we see a shift here but we would like to know the calculation here where the shift balances the force from the external field. Thus, we can look at the force from the electron cloud to the nucleus would be

$$F_{cloud \rightarrow nucleus} = Ze\vec{E}_{cloud} = -Ze\vec{E}_{ext}$$

$$E_{cloud} = \frac{\rho r}{3\epsilon_0} \hat{r}$$

Lecture 13

4.2 Review

Mar 10 10:10

Recall the multipole expansion we had. Note that for dipoles we have the following relationship:

$$\Phi \propto p \propto d$$

$$\Phi \rightarrow \sum A_n \frac{1}{r^n}$$

4.3 Polarization

Note that dielectrics are materials that do not allow current flow. Suppose we had an induced dipole where the proton moves away from the center of the electron cloud by an amount dx for charge $+Ze$. We have the following relationship:

$$\vec{F}_{c \rightarrow n} = Ze\vec{E}_{cloud}(d\vec{x}) = -\vec{F}_{e \rightarrow n}$$

$$\vec{E}_{cloud}(d\vec{x}) = \frac{\rho}{3\epsilon_0} d\vec{x} = -\vec{E}$$

$$\rho = \frac{-Ze}{\frac{4}{3}\pi a^3}$$

The induced dipole moment is therefore going to be

$$\vec{p} = Zedx = 4\pi\epsilon_0 a^3 \vec{E} = \alpha \vec{E}$$

The order of magnitude of the estimation for α is

$$\alpha_H = 4\pi\epsilon_0 (0.53 \cdot 10^{-10})^3 m^3 = 1.7 \times 10^{-41} C^2 \frac{m}{N}$$

The true value differs only slightly where $\alpha_H = 7.3 \times 10^{-41}$. Note that $\vec{p} \equiv \alpha \vec{E}$. In general we have that

$$\vec{p} = \alpha \vec{E}$$

$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & & \\ & \dots & \\ & & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

For the alignment of polar molecules, consider that of H_2O

Example. Suppose we had a H₂O molecule, we know the net forces must cancel on the system

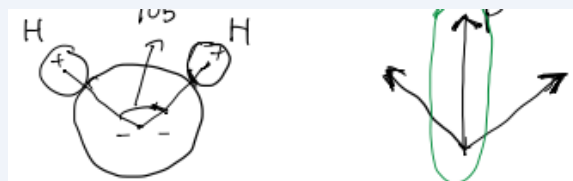


Figure 4.3

The torque then is given as

$$\begin{aligned}\vec{\tau}_0 &= (\vec{r}_+ \times \vec{F}_+) + (\vec{r}_- \times \vec{F}_-) = q(\vec{r}_+ - \vec{r}_-) \times \vec{E} \\ &= q\vec{d} \times \vec{E} = \vec{p} \times \vec{E}\end{aligned}$$

The following holds for non-uniform $\vec{E}$ fields which we take a Taylor series given that d is small

$$\vec{F} = \vec{F}_+ + \vec{F}_- = q(\vec{E}_+ - \vec{E}_-) = (\vec{p} \cdot \vec{\nabla}) \vec{E}$$

For extern electric fields, they allow for an induced dipole moment which helps line up existing dipole moments in the microscopic sense. This leads to a macroscopic dipole $\vec{p}$ polarization.

$$\vec{P}(\vec{r}) \equiv N(\vec{r})\vec{p}(\vec{r})$$

where $\vec{P}$ is the polarization or total dipole movement per unit volume. N is the density of atoms/molecules and $\vec{p}$ is the averaged single atom dipole moment.

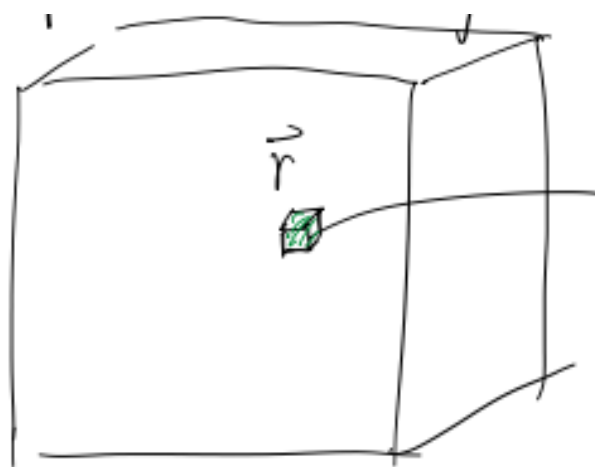


Figure 4.4

For many atoms/molecules, then

$$\vec{p}(\vec{r}) = \vec{p}_{avg} = \frac{1}{n} \sum_{i=1}^n \vec{p}_i$$

Observe if $\vec{E} = 0$ then $\vec{P}_i = 0$ or $\vec{p} \neq 0$? And for $N(\vec{r})$ uniform we simply have $\vec{P}(\vec{r}) = N\vec{p}(\vec{r})$

4.4 Field from a Polarized Object

Considering the potential of a single dipole we have that

$$\Phi_i(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p}_i \cdot (\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3}$$

For a small volume the potential is denoted

$$d\Phi = \frac{1}{4\pi\epsilon_0} \frac{n\vec{p}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

rewrite $n = NdV$

$$d\Phi = \frac{1}{4\pi\epsilon_0} \frac{\vec{P}(\vec{r}') \cdot \vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} dV$$

Therefore the total potential can be written as a volume integral

$$\Phi(\vec{r}) = \int d\Phi = \frac{1}{4\pi\epsilon_0} \int_V \frac{\vec{P}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV'$$

Observe the property

$$\vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Then $\Phi(\vec{r})$ is rewritten

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P}(\vec{r}') \cdot \vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) dV'$$

Using the product rule: $\nabla \cdot (f\vec{A})$ we can rewrite

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\int_V \vec{\nabla} \cdot \left(\frac{\vec{P}(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) dV' - \int_V \frac{1}{|\vec{r} - \vec{r}'|} (\vec{\nabla} \cdot \vec{P}(\vec{r}')) dV' \right]$$

We can rewrite the first integral using Gauss's law and we can convert to the form where the first integral gives

$$\Phi_{surf} = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma ds'}{|\vec{r} - \vec{r}'|}$$

And the second integral gives

$$\Phi_{volume} = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho dV'}{|\vec{r} - \vec{r}'|}$$

We denote that $\sigma_b = \vec{P} \cdot \hat{n}$ as the surface charge density of the bound charge and $\rho_b = -\vec{\nabla} \cdot \vec{P}$ as the volume charge density of the bound charge. Physically, we can represent it as the following diagram:

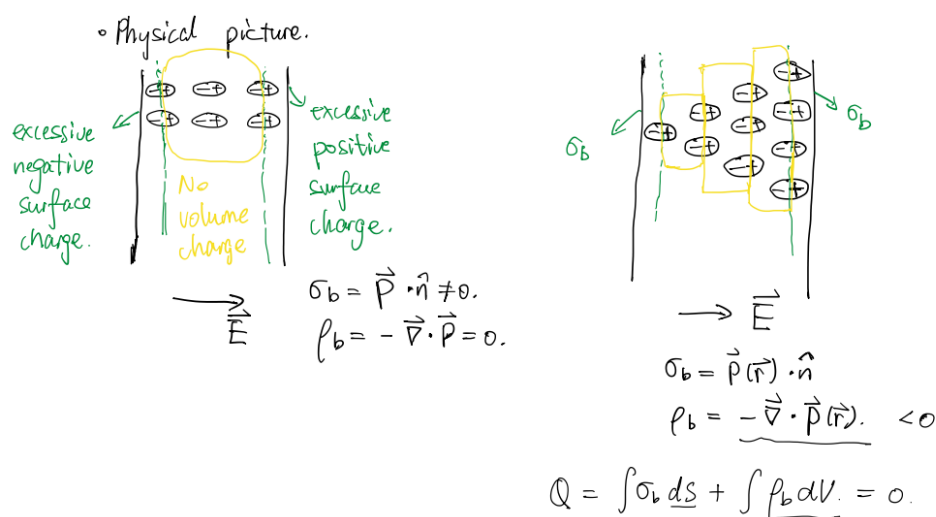


Figure 4.5

Example. Consider the electric field of a uniformly polarized sphere where

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

$$\sigma_b = \vec{P} \cdot \hat{n} = P \cos \theta$$

Using separation of variables and the given B.c. we get that

$$\Phi(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta = \frac{P}{3\epsilon_0} z, & r \leq R; \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta, & r > R; \end{cases}$$

The electric field inside the sphere is going to be

$$\vec{E} = -\vec{\nabla}\Phi = -\frac{P}{3\epsilon_0} \hat{z} = -\frac{\vec{P}}{3\epsilon_0}, \quad r \leq R$$

The total dipole moment of $\vec{p}$ of the sphere is

$$\vec{p} = \vec{P}V = \frac{4}{3}\pi R^3 \vec{P}$$

$$\Phi(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

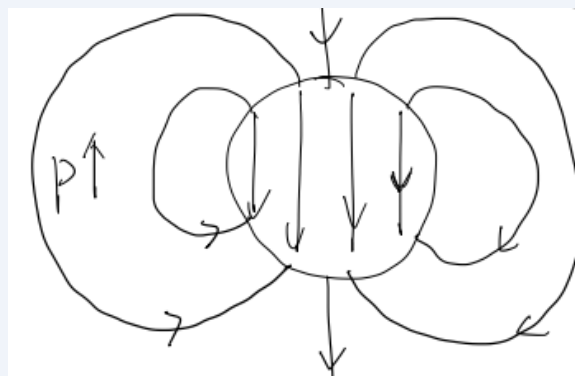


Figure 4.6

Lecture 14

Review: Recall the dielectric response to an external electric field to be that of an induced dipole as well as an alignment of polar molecules leading to a total polarization denoted as $\vec{P}$. Additionally, remember the definition of bound charges where

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$$\sigma_b = \vec{P} \cdot \hat{n} \quad \rho_b = -\nabla \cdot \vec{P}$$

Example. Consider an optical tweezer given by the following diagram:

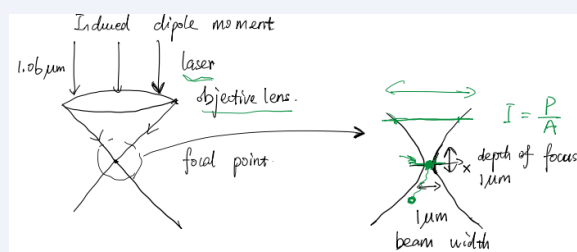


Figure 4.7

The principle of how this device works is that the polarized particles feel a force in the gradient of the laser field. Consider the static electric field where

$$\begin{aligned}\vec{p} &= \alpha \vec{E} \\ \vec{F} &= -\nabla W(\vec{r}) \\ W &= -\int \vec{F} \cdot d\vec{l} = -\int (\vec{p} \cdot \nabla) \vec{E} \cdot d\vec{l} \\ &= -\alpha \int (\vec{E} \cdot \nabla) \vec{E} \cdot d\vec{l}\end{aligned}$$

Using a vector identity we have

$$= -\frac{\alpha}{2} \int \nabla (\vec{E} \cdot \vec{E}) \cdot d\vec{l} = -\frac{\alpha}{2} |\vec{E}|^2$$

Given that $\vec{E} = \vec{E}_0 \cos \omega t$. The particle moves with the oscillations. Taking the time average of the force we do this from the work where

$$W = -\frac{\alpha}{2} \langle \vec{E} \cdot \vec{E} \rangle$$

$$W = -\frac{\alpha}{4} |E_0(\vec{r})|^2$$

We can write this in terms of the intensity $I \equiv \frac{1}{2} \epsilon_0 c E_0^2$

$$W = -\frac{\alpha}{2\epsilon_0 c} I(\vec{r})$$

$$\vec{F} = -\nabla W = \frac{\alpha}{2\epsilon_0 c} \nabla I$$

Taking the peak we have

$$\begin{aligned}|\nabla I|_\infty &= \frac{P}{\pi d^2} \\ \vec{F} &= \frac{\alpha}{2\pi\epsilon_0 c} \frac{P}{d^3}\end{aligned}$$

4.5 Dielectric Constant and Susceptibility

We have the relationship that $\vec{P} \propto \vec{E}$

$$\vec{P} = \epsilon_0 \chi_e \vec{E} = \epsilon_0 \epsilon_r \vec{E}, \quad \epsilon_r = 1 + \chi_e$$

ϵ_r is the relative permittivity or the dielectric constant. We can therefore write that

$$\epsilon = \epsilon_0 \epsilon_r = \epsilon_0 (1 + \chi_e)$$

For a homogeneous dielectric, we must have that

$$\begin{aligned}\rho_b &= -\nabla \cdot \vec{P} = -\nabla \cdot (\epsilon_0 \chi_e \vec{E}) \\ &= -\frac{\epsilon_0 \chi_0}{\epsilon} \rho_f\end{aligned}$$

Using this concept, consider an example a external electric field and given polarization with a hole in the middle with no $\vec{E}$ field. Therefore we would have that

$$\begin{aligned}\vec{E}_L &= \vec{E}_{ext} + \vec{E}_{other} = \vec{E}_{ext} + \vec{E}_{P_{total}} - \vec{E}_{sph} = \vec{E} - \left(-\frac{\vec{p}}{3\epsilon_0}\right) \\ \vec{p} &= \alpha \vec{E}_2 = \alpha \left(\vec{E} + \frac{\vec{p}}{3\epsilon_0}\right)\end{aligned}$$

$$\vec{P} = N\vec{p} = N\alpha \left(\vec{E} + \frac{\vec{p}}{3\epsilon_0} \right)$$

Rewriting we have

$$\chi_0 = \frac{N \frac{\alpha}{\epsilon_0}}{1 - N \frac{\alpha}{3\epsilon_0}}$$

$$\alpha = \frac{3\epsilon_0}{N} \left(\frac{\chi}{\chi + 3} \right) = \frac{3\epsilon_0}{N} \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right)$$

Lecture 15

4.5.1 Review

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Recall the electric displacement field which is given as

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

and the Gauss's law give as

$$\nabla \cdot \vec{D} = \rho_f$$

There are more concepts on how the dielectric reacts to an external field.

1. Dielectric susceptibility is the amount of polarization in a given electric field given as $\vec{P} = \epsilon_0 \chi \vec{E}$
2. The relative permittivity is given as $\epsilon = 1 + \chi_e$
3. We built a relationship between polarizability and susceptibility which is given as $\vec{p} = \alpha \vec{E}$. This takes into account the local field there

$$\alpha = \frac{3\epsilon_0}{N} \frac{\chi_e}{\chi_e + 3}$$

4.6 Temperature Dependence of $\vec{P}$ and ϵ_r

If we have finite temperature, there exists thermal excitations where the molecules will bounce around s.t. the direction of the field is not w.r.t the external field. For a polar molecule, the energy is given as

$$W = -\vec{p}_0 \cdot \vec{E}$$

From statistical mechanics, the probability of finding a particle in an ensemble with an energy is given as

$$P \propto e^{-\frac{W}{k_B T}}$$

where k_B is given as Boltzman constant and T is given as the absolute temperature. Therefore we write

$$P(W = W_0) \propto \exp -\frac{W_0}{k_B T}$$

Given the intrinsic dipole moments are the same for different molecules we have that the expectation value of $\langle p \rangle$, the ensemble average, is given as

$$\langle p \rangle = \langle p_0 \cos \theta \rangle = \frac{1}{\int_0^\pi \exp\left(-\frac{p_0 E \cos \theta}{k_B T}\right) \sin \theta d\theta} \int_0^\pi p_0 \cos \theta \exp\left(-\frac{p_0 E \cos \theta}{k_B T}\right) \sin \theta d\theta$$

Let $x = \frac{p_0 E \cos \theta}{k_B T} \Rightarrow dx = -\frac{p_0 E}{k_B T} \sin \theta d\theta$. Rewriting we have

$$\begin{aligned} \langle p \rangle &= \frac{k_B T}{E} \frac{\int_{-x_0}^{x_0} x e^x dx}{\int_{-x_0}^{x_0} e^x dx} \\ &= \frac{k_B T}{E} (c_0 \coth x_0 - 1) \end{aligned}$$

$$= p_0 \coth\left(\frac{p_0 E}{k_B T}\right) - \frac{k_B T}{E}$$

We can now write for the polarization of the whole system as

$$P = N \langle p \rangle N \left[p_0 \coth\left(\frac{p_0 E}{k_B T}\right) - \frac{k_B T}{E} \right]$$

If plotted against the external field, it will eventually saturate for large fields.

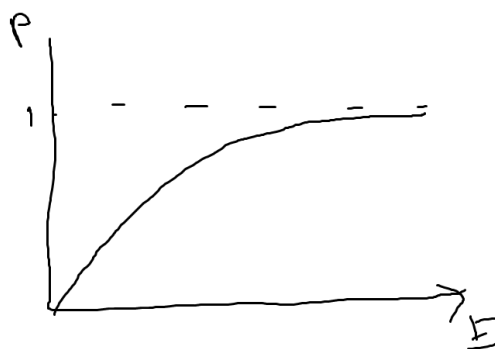


Figure 4.8

For small $\vec{E}$ where $\frac{p_0 E}{k_B T} \ll 1$, the polarization is given as

$$P \approx N p_0 \frac{1}{3} \frac{p_0 E}{k_B T} = \frac{N p^2 E}{3 k_B T}$$

4.7 Electrostatic energy in Dielectrics

Suppose we have a dielectric with a free charge at the bottom with ρ_f

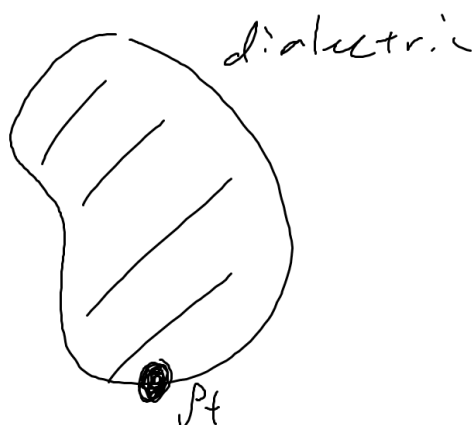


Figure 4.9

The work required to bring in the free charge ρ_f is given as

$$\Delta W = \int (\Delta \rho_f) \Phi dV$$

$$lr_f = \vec{\nabla} \cdot \vec{D} \quad \Delta\rho_f = \vec{\nabla} \cdot (\Delta\vec{D})$$

Therefore we can write the work as

$$\Delta W = \int \nabla \cdot (\Delta\vec{D}) \Phi dV$$

Using the chain rule

$$\begin{aligned} &= \int \nabla \cdot (\Delta\vec{D}\Phi) dV - \int (\Delta\vec{D} \cdot \nabla\Phi) dV \\ &= \oint_S \Phi \Delta\vec{D} \cdot \hat{n} ds + \int (\Delta\vec{D} \cdot \vec{E}) dV \end{aligned}$$

The first term has it such that at best for monopoles the potential falls as $\frac{1}{r}$ and the displacement field falls as $\frac{1}{r^2}$. The surface area grows as r^2 . Thus, the integral grows as $\frac{1}{r}$. When we look at the whole space the first term must go to 0. Thus the first integral must be 0.

The second term can be rewritten to be

$$= \int (\epsilon \Delta\vec{E} \cdot \vec{E}) dV$$

where both terms ϵ and E are functions of position. We can rewrite this as

$$\Delta W = \frac{1}{2} \int \epsilon \Delta (\vec{E} \cdot \vec{E}) dV = \frac{1}{2} \int \Delta (\vec{D} \cdot \vec{E}) dV$$

This is work done to increase the density of the free charge. Thus, the total work to move the free charge to its position is given by

$$W = \frac{1}{2} \int \vec{D} \cdot \vec{E} dV$$

Example. Consider a parallel plate capacitor w/ dielectric between them where $Q = \pm\sigma A$ with an $\vec{E}$ pointing up. Without a dielectric, the $\vec{E}$ field is given as

$$\vec{E}_0 = \frac{\sigma}{\epsilon_0} \hat{y}$$

$$\sigma_b = \vec{P} \cdot \hat{n} = P = (\epsilon - \epsilon_0) E_{in}$$

$$\rho_b = \nabla \cdot \vec{P}$$

We will have that

$$\vec{E}_b = -\frac{\sigma_b}{\epsilon_0} \hat{y} = -\left(\frac{\epsilon}{\epsilon_0} - 1\right) \vec{E}_{in}$$

We know that the electric field inside is given as

$$\vec{E}_{in} = \vec{E}_0 + \vec{E}_b = \frac{\sigma}{\epsilon_0} \hat{y} - \left(\frac{\epsilon}{\epsilon_0} - 1\right) \vec{E}_{in}$$

Solving as a function of σ ;

$$\vec{E}_{in} = \frac{\sigma}{\epsilon} \hat{y} = \frac{\sigma}{\epsilon_0 \epsilon_r} \hat{y}$$

From here we can write the potential

$$\Phi = E_{in} d = \frac{\sigma d}{\epsilon_0 \epsilon_r} = \frac{Qd}{A \epsilon_0 \epsilon_r}$$

The capacitance is written as

$$C = \frac{Q}{\Phi} = \frac{\epsilon_r \epsilon_0 A}{d = \epsilon_r c_0}$$

The energy stored is given as

$$W = \frac{1}{2} Q \Phi = \frac{Q^2 d}{2 \epsilon_0 \epsilon_r A} = \frac{W_0}{\epsilon_r}$$

We see if we fix Φ , then

$$Q = \frac{\Phi \epsilon_r \epsilon_0 A}{d}$$

$$W = \frac{1}{2} Q \Phi = \frac{\Phi^2 \epsilon_0 \epsilon_r A}{2d} = \epsilon_r W_0$$

thus we see an enhancement by ϵ_r , Thus, the dielectric material changes the situation.

Example. Consider another parallel plate capacitor with a dielectric s.t. it is slightly in the capacitor. Thus, it is not inside by a distance x . Suppose we fix Q . Then we have that

$$F = -\frac{dW}{dx}$$

We can rewrite this simply as a sum of two capacitors C_1 and C_2 where the total capacitance is written as

$$C = C_1 + C_2 = \frac{A_1 \epsilon_0}{d} + \frac{A_2 \epsilon_0 \epsilon_r}{d}$$

$$\frac{a \epsilon_0}{d} [x + \epsilon_r (w - x)]$$

$$= \frac{a \epsilon_0}{d} [x(1 - \epsilon_r) + \epsilon_r w]$$

$$W = \frac{Q^2}{2C} = \frac{1}{2} \frac{Q^2 d}{a \epsilon_0 [\epsilon_r w - (\epsilon_r - 1)x]}$$

Thus we have a relationship between x and W . Therefore we get that

$$F = -\frac{dW}{dx} = -\frac{1}{2} \frac{Q^2 d}{a \epsilon_0} \frac{\epsilon_0 - 1}{[\epsilon_r w - (\epsilon_0 - 1)x]^2}$$

Thus, we see the force pulls the slab back into place. We can additionally think of the problem if we fix the potential instead and is left as an exercise. Note that if we ignore the edge effects present, the capacitance would have a uniform field with different fields in the dielectric and vacuum region. This doesn't push it in or out. The fringe field is giving an overall force where there are curved field lines. The induced/bound charge on the dielectric (fringe field) puts a force on the bound charge which moves it. Another way to think of it is that the charges of the metal plate generates a force. Note the fringe field is complicated to calculate, we can know the fringe field force by knowing the energy without calculating the fringe field.